The Fabius Function and Rvachev's Up-Function

Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics

Human-readable synthesis of the formal development

Vladimir Reshetnikov's `ProveIt` repository

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Abstract

The Fabius function is the unique smooth distribution function that is constant outside $[0, 1]$, symmetric about $1/2$, and satisfies the self-similar differential equation $F'(x) = 2F(2x)$ on the left half of the unit interval. Folding it about the midpoint gives Rvachev's compactly supported even function up, one of the canonical examples of a C^∞ function with compact support that is given by an exact refinement equation. The same object admits several sharply different descriptions: a contraction fixed point, a probability law of an infinite weighted sum of independent uniform random variables, an infinite convolution, an infinite product of sinc functions, a Thue–Morse translate sum, and a family of exact rational recurrences.

This article develops these descriptions and proves their equivalence. Particular emphasis is placed on two subjects for which the formal development goes substantially beyond the usual short accounts. First, every dyadic value is evaluated by an explicit finite Thue–Morse/moment formula, and also by a fast terminating recursion that removes one set bit of the numerator at each step. Second, the small-argument asymptotic is derived by an exact negative-Laplace decomposition and a quantitative saddle-point argument. The correct main term contains a genuine nonconstant periodic fluctuation in the lower Lambert coordinate. Its Fourier coefficients are explicit Gamma–zeta values. Omitting that fluctuation, as in a frequently reproduced elementary formula, destroys the claimed inverse-logarithmic error bound. The first saddle correction and the full all-orders expansion are also described.

The presentation follows the theorem structure of the Lean formalization under `Analysis/FabiusFunction`, while translating the machine-oriented decomposition into conventional mathematical arguments. Source-level endpoint and normalization defects are corrected rather than repeated.

Contents

1 Historical note and scope

The probability distribution now attached to the name of Fabius is older than Fabius's two-page note. It occurs in the work of Jessen and Wintner on infinite convolutions, was rediscovered several times, and entered a different mathematical culture through the compactly supported "finite function" of V. A. Rvachev. Fabius used it in 1966 as a particularly economical probabilistic construction of a nowhere analytic C^∞ function [2]. Rvachev's normalization emphasizes the even compactly supported density, whereas the Fabius normalization emphasizes its cumulative distribution function. Arias de Reyna's 1982 paper gave a systematic treatment of the compactly supported function, including its Fourier product, probability interpretation, partitions of unity, derivatives, and dyadic values; an English translation appeared in 2017 [4]. His later arithmetic paper organized the exact rational sequences and denominator questions [5].

The present article is not merely a paraphrase of those papers. It follows the proof dependency graph of the Lean development in the `ProveIt` repository [7]. That development first constructs the bounded distribution function, then derives the Rvachev and signed-global forms, and finally builds the arithmetic and asymptotic layers on top. This order has two advantages. It prevents identities valid for the signed extension from being silently applied to the bounded distribution, and it makes the exact dyadic algorithms independent of any numerical approximation.

The formalization also settles several questions that are only implicit, conjectural, or incorrectly normalized in the source material. Most importantly, the sharp endpoint asymptotic contains a nonconstant periodic function of an exact lower-Lambert phase. The periodic term is not a cosmetic improvement: every nonzero Fourier mode is explicit and nonzero, and an elementary formula without it fails at the error scale usually claimed. The development further proves a full inverse-Lambert expansion, establishes an exact bit-recursive evaluator for every dyadic rational, and supplies an absolutely and uniformly convergent Fourier–Legendre expansion of up.

Our convention is fixed throughout:

- F is the bounded cumulative distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported density on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$.

All logarithms are natural logarithms. We write $L = \log 2$ in the asymptotic sections, $\mathrm{wt}_2(n)$ for the sum of the binary digits of n, and $\operatorname{sinc} z = \sin z/z$ with the removable value 1 at the origin.

2 Three functions that must not be conflated

The literature often uses the same letter F for three related, but globally different, functions. That economy of notation is harmless in a local calculation and dangerous in a long proof. We therefore begin by separating them.

2.1 The bounded Fabius distribution function

Definition 2.1 (Bounded Fabius function)**.** A function $F : \mathbb{R} \to [0, 1]$ is called a *bounded Fabius function* if

$$
\begin{aligned}
F(x) &= 0 && (x \le 0), && (1)\\
F(x) &= 1 && (x \ge 1), && (2)\\
F(1-x) &= 1 - F(x) && (0 \le x \le 1), && (3)\\
F'(x) &= 2F(2x) && (0 \le x \le \tfrac{1}{2}), && (4)
\end{aligned}
$$

and F is infinitely differentiable on $\mathbb{R}$.

The derivative equation on the right half follows from symmetry:

$$F'(x) = 2F(2-2x), \qquad \tfrac{1}{2} \le x \le 1. \tag{5}$$

In particular $F'(x) > 0$ for $0 < x < 1$, once positivity in the interior is known. Thus F is a genuine cumulative distribution function, not merely a smooth interpolation between 0 and 1.

Because F is constant on both exterior half-lines and is C^∞, every derivative of positive order vanishes at both endpoints:

$$F^{(m)}(0) = F^{(m)}(1) = 0 \qquad (m \ge 1). \tag{6}$$

This flatness is the source of both its extraordinary endpoint decay and its failure to be analytic there.

2.2 Rvachev's compactly supported bump

Definition 2.2 (Rvachev up-function)**.** Given F, define

$$\mathrm{up}(x) = \begin{cases} F(x+1), & x \le 0, \\ F(1-x), & x > 0. \end{cases} \tag{7}$$

Equivalently,

$$\mathrm{up}(x) = F(1-|x|). \tag{8}$$

Here the right side is interpreted using the globally bounded function F.

It follows immediately that up is even, nonnegative, supported on $[-1,1]$, and $\mathrm{up}(0) = 1$. On the unit interval the two functions are simply translates:

$$F(x) = \mathrm{up}(x-1), \qquad 0 \le x \le 1. \tag{9}$$

The graph of F is an infinitely flat sigmoid; the graph of up is its symmetric fold.

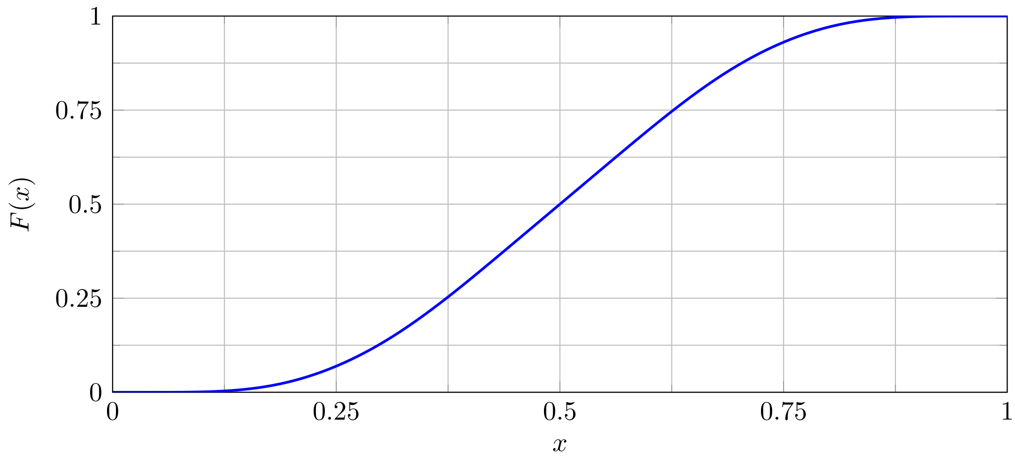

Figure 1: The bounded Fabius function on $[0,1]$, drawn through exact dyadic values on a 2^{-8} grid. At this resolution the straight segments are visually indistinguishable from the smooth curve.

2.3 The signed global extension

Let $\mathrm{wt}_2(n)$ denote the sum of the binary digits of n, and put

$$\varepsilon_n = (-1)^{\mathrm{wt}_2(n)}. \tag{10}$$

The sequence (ε_n) is the $\{\pm 1\}$ form of the Thue–Morse sequence.

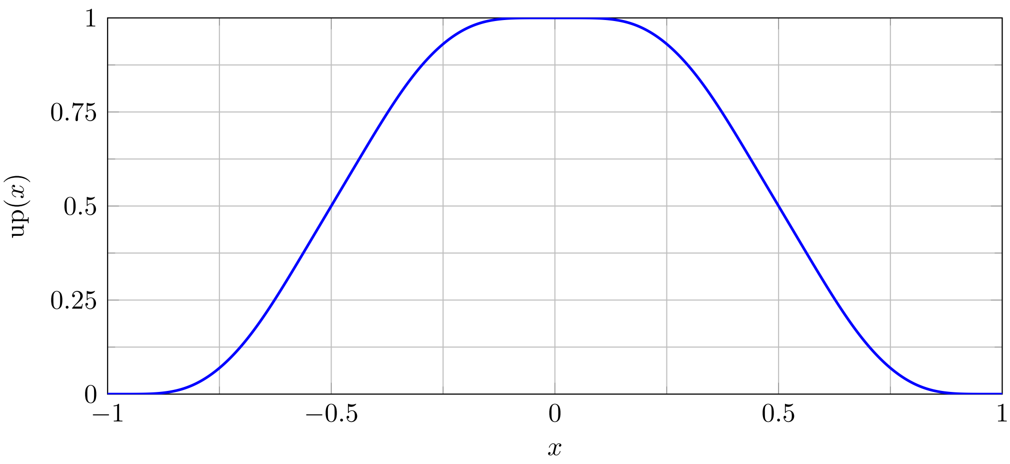

Figure 2: Rvachev's up-function on its support $[-1,1]$, obtained by folding the same exact dyadic data.

Definition 2.3 (Signed extension)**.** Define

$$\mathcal{F}(x) = \sum_{n\geq 0} \varepsilon_n \ \mathrm{up}(x-2n-1). \tag{11}$$

At a fixed x only finitely many summands can be nonzero, so this is a locally finite sum. On every block $[2b, 2b+2]$ there is exactly one nonzero translate:

$$\mathcal{F}(x) = \varepsilon_b \, \mathrm{up}(x-2b-1), \qquad 2b \leq x \leq 2b+2. \tag{12}$$

It vanishes for $x \leq 0$, and it agrees with the bounded F on $[0,1]$. It does *not* remain in $[0,1]$ to the right of that interval. This is the global object that satisfies the cleanest differential equation:

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \qquad (x \in \mathbb{R}). \tag{13}$$

Arias de Reyna's arithmetic paper uses this extension. Statements about its values outside $[0,1]$ must therefore not be transferred blindly to the bounded cumulative distribution.

Notation used in this article. F always means the bounded distribution function; up always means the compactly supported even bump; and $\mathcal{F}$ always means the signed global extension. On $[0,1]$ we have $F = \mathcal{F}$, but globally they are different functions.

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3 Existence and uniqueness by a contraction

The most economical construction of F is a Banach fixed-point argument. It has an additional advantage: the contraction estimate later gives a clean uniqueness theorem, without presupposing Fourier analysis or probability.

3.1 The admissible function space

Let $I = [0,1]$ and let $C(I)$ carry the supremum norm. Define $\mathcal{A} \subset C(I)$ by

$$\mathcal{A} = \{f \in C(I) : 0 \leq f \leq 1, \ f(1-x) = 1 - f(x)\}. \tag{14}$$

The set $\mathcal{A}$ is closed in the Banach space $C(I)$, hence complete. It is nonempty; for example $f(x) = x$ belongs to it.

Extend $f \in \mathcal{A}$ to a continuous function $\widetilde{f}$ on $\mathbb{R}$ by clamping the argument to I:

$$\widetilde{f}(t) = \begin{cases} f(0), & t \le 0, \\ f(t), & 0 \le t \le 1, \\ f(1), & t \ge 1. \end{cases} \tag{15}$$

Symmetry gives

$$\int_0^1 f(t)\,\mathrm{d}t = \frac{1}{2}. \tag{16}$$

Indeed, replacing t by $1-t$ and adding the two integrals gives $2\int_0^1 f = 1$.

3.2 The integral transform

For $f \in \mathcal{A}$, define Tf by

$$(Tf)(x) = \begin{cases} \displaystyle\int_0^{2x} \widetilde{f}(t)\,\mathrm{d}t, & 0 \le x \le \frac{1}{2}, \\ 1 - \displaystyle\int_0^{2-2x} \widetilde{f}(t)\,\mathrm{d}t, & \frac{1}{2} < x \le 1. \end{cases} \tag{17}$$

The two definitions agree at $x = 1/2$ by (16). Positivity and the upper bound $f \le 1$ show that $0 \le Tf \le 1$, and a direct substitution shows $(Tf)(1-x) = 1 - (Tf)(x)$. Hence $T : \mathcal{A} \to \mathcal{A}$.

The central estimate is stronger than the naive integral bound.

Lemma 3.1 (Contraction estimate)**.** *For $f, g \in \mathcal{A}$,*

$$\|Tf - Tg\|_\infty \le \frac{1}{2}\|f - g\|_\infty. \tag{18}$$

Proof. Write $h = f - g$. Since both f and g have integral $1/2$, $\int_0^1 h = 0$. For $0 \le y \le 1$ we may therefore estimate the primitive in two ways:

$$\left|\int_0^y h(t)\,\mathrm{d}t\right| \le y\|h\|_\infty,$$
$$\left|\int_0^y h(t)\,\mathrm{d}t\right| = \left|\int_y^1 h(t)\,\mathrm{d}t\right| \le (1-y)\|h\|_\infty.$$

Consequently

$$\left|\int_0^y h(t)\,\mathrm{d}t\right| \le \min(y, 1-y)\|h\|_\infty \le \frac{1}{2}\|h\|_\infty.$$

Every value of $Tf - Tg$ is one of these primitives, possibly with a minus sign. Taking the supremum proves (18). □

Theorem 3.2 (Existence of the bounded Fabius function)**.** *The transform T has a unique fixed point $f_* \in \mathcal{A}$. Extending f_* by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$ produces the bounded Fabius function F.*

Proof. The space $\mathcal{A}$ is complete and T is a strict contraction with constant $1/2$, so Banach's fixed-point theorem gives a unique fixed point. On $[0, 1/2]$ its fixed-point equation is

$$F(x) = \int_0^{2x} F(t)\,\mathrm{d}t. \tag{19}$$

Differentiating gives $F'(x) = 2F(2x)$. The second half of (17) supplies symmetry and the matching right-hand formula. The endpoint values are immediate. □

Starting from any admissible f_0, the iterates $f_{n+1} = Tf_n$ converge uniformly to F, with the explicit estimate

$$\|f_n - F\|_\infty \le 2^{-n}\|f_0 - F\|_\infty. \tag{20}$$

This gives a conceptually simple approximation scheme, although exact dyadic evaluation will be much faster.

3.3 The differential equation for up and smoothness

Folding (4) and (5) yields the refinement-type differential equation

$$\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big) \qquad (x \in \mathbb{R}). \tag{21}$$

At $x = 0$ the two one-sided derivatives glue because $\mathrm{up}(1) = \mathrm{up}(-1) = 0$. Once up is continuously differentiable, (21) bootstraps regularity: if $\mathrm{up} \in C^r$, then the right side is C^r, so $\mathrm{up} \in C^{r+1}$. Induction gives $\mathrm{up} \in C^\infty$ and then $F \in C^\infty$.

Proposition 3.3 (Basic shape). *The functions F and* up *have the following properties.*

1. *F is strictly increasing on $(0,1)$ and satisfies $0 < F(x) < 1$ there.*
2. up *is even, strictly positive on $(-1,1)$, and zero outside $[-1,1]$.*
3. $\int_{-1}^{1} \mathrm{up}(x)\,\mathrm{d}x = 1$.
4. *Every derivative of* up *vanishes at $x = \pm 1$.*

Proof. For $0 < x \le 1/2$, (4) gives $F'(x) = 2F(2x)$. Positivity follows first near $1/2$ from $F(1/2) = 1/2$, and then propagates toward zero by repeatedly doubling the argument; symmetry handles the right half. This proves strict monotonicity and, by (8), positivity of up in the interior of its support. Finally,

$$\int_{-1}^{1} \mathrm{up}(x)\,\mathrm{d}x = 2\int_0^1 F(t)\,\mathrm{d}t = 1,$$

where the middle identity uses the fold and the last uses symmetry. Flatness at the support endpoints follows from smoothness and the fact that up is identically zero just outside the support. □

3.4 The original Rvachev characterization

The preceding construction is equivalent to the characterization historically associated with Rvachev.

Theorem 3.4 (Original characterization). *There is a unique nonzero C^∞ function $\varphi : \mathbb{R} \to \mathbb{R}$ such that*

$$\operatorname{supp}\varphi = [-1,1], \qquad \int_{\mathbb{R}} \varphi(x)\,\mathrm{d}x = 1, \tag{22}$$

and

$$\varphi'(x) = k\big(\varphi(2x+1) - \varphi(2x-1)\big) \tag{23}$$

for some real scale k. Necessarily $k = 2$, and $\varphi = \mathrm{up}$.

Proof architecture. Integrating (23) against suitable test functions, or taking its Fourier transform, first forces $k = 2$ under the normalization (22). With the Fourier convention

$$\widehat{\varphi}(z) = \int_{\mathbb{R}} \varphi(x)\mathrm{e}^{-2\pi \mathrm{i} xz}\,\mathrm{d}x, \tag{24}$$

one obtains

$$\widehat{\varphi}(z) = \operatorname{sinc}(\pi z)\widehat{\varphi}(z/2), \qquad \operatorname{sinc} z = \begin{cases} \sin z/z, & z \ne 0, \\ 1, & z = 0. \end{cases} \tag{25}$$

Iterating N times gives a finite sinc product multiplied by $\widehat{\varphi}(z/2^N)$. Continuity and $\widehat{\varphi}(0) = 1$ determine the Fourier transform uniquely in the limit. Fourier inversion then determines φ. The function constructed above satisfies all the hypotheses, so it is that unique solution. □

4 The probability law behind the functions

The fixed-point construction is analytic. A probabilistic construction explains the same self-similarity almost without calculation and makes monotonicity and positivity transparent.

4.1 An infinite weighted sum of uniforms

Let $U_0, U_1, \ldots$ be independent random variables, each uniformly distributed on $[0, 1]$, and define

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \tag{26}$$

The series converges absolutely for every outcome and satisfies $0 \le X \le 1$. Removing the first coordinate gives an independent copy X' of X, and

$$X = \frac{U_0 + X'}{2}. \tag{27}$$

Reflecting every coordinate, $U_n \mapsto 1 - U_n$, preserves the product measure and sends X to $1 - X$.

Let $H(x) = \mathbb{P}(X \le x)$ be the cumulative distribution function. Conditioning on U_0 in (27) gives

$$H(y) = \int_0^1 H(2y - u)\,\mathrm{d}u. \tag{28}$$

For $0 \le y \le 1/2$, the integrand vanishes when $u > 2y$, so

$$H(y) = \int_0^{2y} H(t)\,\mathrm{d}t. \tag{29}$$

Reflection gives $H(1 - y) = 1 - H(y)$. Thus the restriction of H to $[0, 1]$ is a fixed point of the contraction T from section 3.

Theorem 4.1 (Probabilistic representation). *For every $x \in \mathbb{R}$,*

$$F(x) = \mathbb{P}\left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \le x\right). \tag{30}$$

For $-1 \le y \le 0$,

$$\mathrm{up}(y) = \mathbb{P}(X \le y + 1). \tag{31}$$

Proof. The preceding calculation shows that H is an admissible fixed point of T. Uniqueness of the fixed point gives $H = F$. Equation (31) is then just the left branch of (7). □

The more symmetric random variable is

$$Y = 2X - 1 = \sum_{n=0}^{\infty} \frac{V_n}{2^{n+1}}, \qquad V_n = 2U_n - 1 \sim \mathrm{Unif}[-1, 1]. \tag{32}$$

The density of Y is precisely up. Indeed, for $-1 < y < 1$,

$$f_Y(y) = \frac{1}{2} F'\left(\frac{y+1}{2}\right) = \mathrm{up}(y). \tag{33}$$

This identity is a compact way to remember why up integrates to one.

4.2 Finite convolutions and step approximants

For each n, the summand $V_n/2^{n+1}$ is uniform on $[-2^{-n-1}, 2^{-n-1}]$ and has density

$$b_n(x) = 2^n\, \mathbf{1}_{[-2^{-n-1},2^{-n-1}]}(x). \tag{34}$$

Hence the partial sum

$$Y_N = \sum_{n=0}^{N} \frac{V_n}{2^{n+1}} \tag{35}$$

has density

$$u_N = b_0 * b_1 * \cdots * b_N. \tag{36}$$

These compactly supported piecewise-polynomial densities converge to up. Weak convergence is immediate from almost-sure convergence of Y_N; the refinement equation and uniform bounds upgrade the convergence to the strong forms used in the formal development. This constructs up as an infinite convolution of shrinking boxes, a perspective useful in approximation theory and wavelet analysis.

5 Fourier transform, entire products, and Poisson summation

5.1 The sinc product

We use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i} x z}\, \mathrm{d}x. \tag{37}$$

The characteristic function of a uniform variable on $[-a, a]$ is $\operatorname{sinc}(2\pi a z)$. Independence in (32) therefore yields the product formula.

Theorem 5.1 (Fourier product)**.** *For every $z \in \mathbb{C}$,*

$$\widehat{\mathrm{up}}(z) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right). \tag{38}$$

The product converges locally uniformly and hence defines an entire function.

Proof. For real z, the characteristic function of the partial sum Y_N is the finite product through $n = N$. Since $Y_N \to Y$ almost surely and the exponential has modulus one, dominated convergence gives the real identity. On compact subsets of $\mathbb{C}$,

$$\operatorname{sinc} w = 1 - \frac{w^2}{6} + \mathcal{O}(w^4),$$

so the sum of the deviations of the factors from 1 is dominated by $C_K \sum 4^{-n}$. The Weierstrass product theorem gives local uniform convergence to an entire function. Equality on the real axis and the identity theorem extend the formula to all complex z. □

Separating the first factor gives the refinement equation already seen in (25):

$$\widehat{\mathrm{up}}(z) = \operatorname{sinc}(\pi z)\widehat{\mathrm{up}}(z/2). \tag{39}$$

Conversely, continuity at zero, normalization $\widehat{\mathrm{up}}(0) = 1$, and (39) force (38) by iteration. This is the core of the Fourier uniqueness proof.

5.2 Rapid decay and inversion

Since $\mathrm{up} \in C_c^\infty(\mathbb{R})$, repeated integration by parts gives, for every integer $N \geq 0$,

$$|\widehat{\mathrm{up}}(\xi)| \leq C_N(1+|\xi|)^{-N}. \tag{40}$$

Thus the Fourier product is rapidly decreasing on the real axis despite being entire of finite exponential type. Fourier inversion is absolutely convergent:

$$\mathrm{up}(x) = \int_{\mathbb{R}} \widehat{\mathrm{up}}(\xi)\mathrm{e}^{2\pi \mathrm{i} x\xi}\,\mathrm{d}\xi. \tag{41}$$

5.3 Poisson summation and partitions of unity

Rapid decay on both sides makes Poisson summation available without distributional qualifications.

Theorem 5.2 (Poisson summation)**.** *For $u > 0$ and $t \in \mathbb{R}$,*

$$\sum_{k\in\mathbb{Z}} \mathrm{up}(t+uk) = \frac{1}{u}\sum_{k\in\mathbb{Z}} \widehat{\mathrm{up}}(k/u)\mathrm{e}^{2\pi \mathrm{i} kt/u}. \tag{42}$$

Both series converge absolutely and locally uniformly.

For a positive integer m, every nonzero term $\widehat{\mathrm{up}}(mk)$ vanishes because the first sinc factor in (38) vanishes at nonzero integers. Taking $u = 1/m$ in (42) gives the exact partition of unity

$$\sum_{k\in\mathbb{Z}} \mathrm{up}\left(t+\frac{k}{m}\right) = m. \tag{43}$$

In particular,

$$\sum_{k\in\mathbb{Z}} \mathrm{up}(t+k) = 1. \tag{44}$$

This identity is one reason up is useful as a smooth compactly supported averaging kernel.

5.4 Moment expansion of the Fourier transform

Define the even moments

$$c_n = \int_{-1}^{1} x^{2n}\,\mathrm{up}(x)\,\mathrm{d}x. \tag{45}$$

Evenness eliminates the odd moments, and expansion of the exponential in (37) gives the entire series

$$\widehat{\mathrm{up}}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!}(2\pi z)^{2n}. \tag{46}$$

Comparing this with the sinc refinement will produce the exact rational recurrence for c_n in section 7.

6 The global extension and exact derivative identities

The signed extension $\mathcal{F}$ is not merely a device for shortening notation. It turns the piecewise differential structure of F and up into a global scaling law, and from that law one can read off all higher derivatives.

6.1 Thue–Morse splitting

The binary weight satisfies

$$\mathrm{wt}_2(2n) = \mathrm{wt}_2(n), \qquad \mathrm{wt}_2(2n+1) = \mathrm{wt}_2(n) + 1. \tag{47}$$

Therefore

$$\varepsilon_{2n} = \varepsilon_n, \qquad \varepsilon_{2n+1} = -\varepsilon_n. \tag{48}$$

Differentiate the locally finite sum (11) term by term and use (21). The result is a sum over even and odd indices. Reindexing those two subseries with (48) makes them recombine into $2\mathcal{F}(2x)$.

Theorem 6.1 (Global differential equation)**.** *The signed extension is infinitely differentiable and*

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \tag{49}$$

for every real x*.*

Proof. Local finiteness permits differentiation of (11). By (21),

$$\begin{aligned}
\mathcal{F}'(x) &= 2\sum_{n\ge 0} \varepsilon_n \big[\mathrm{up}(2x-4n-1) - \mathrm{up}(2x-4n-3)\big] \\
&= 2\sum_{n\ge 0} \varepsilon_{2n}\,\mathrm{up}(2x-2(2n)-1) + 2\sum_{n\ge 0} \varepsilon_{2n+1}\,\mathrm{up}(2x-2(2n+1)-1) \\
&= 2\sum_{m\ge 0} \varepsilon_m\,\mathrm{up}(2x-2m-1) = 2\mathcal{F}(2x).
\end{aligned}$$

The same local-finiteness argument shows that the sum is C^∞ because each translate of up is. □

6.2 All iterated derivatives

Repeated differentiation of (49) gives a triangular power of two.

Theorem 6.2 (Derivative scaling)**.** *For every integer* $m \ge 0$*,*

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}}\mathcal{F}(2^m x). \tag{50}$$

For $-1 \le x \le 1$*,*

$$\mathrm{up}^{(m)}(x) = 2^{\binom{m+1}{2}}\mathcal{F}(2^m x + 2^m). \tag{51}$$

Proof. The first assertion is immediate by induction. If it holds for m, then

$$\begin{aligned}
\mathcal{F}^{(m+1)}(x) &= 2^{\binom{m+1}{2}} 2^m \mathcal{F}'(2^m x) \\
&= 2^{\binom{m+1}{2}} 2^m \cdot 2\mathcal{F}(2^{m+1}x) \\
&= 2^{\binom{m+2}{2}}\mathcal{F}(2^{m+1}x).
\end{aligned}$$

On $[-1,1]$, $\mathrm{up}(x) = \mathcal{F}(x+1)$, so $\mathrm{up}^{(m)}(x) = \mathcal{F}^{(m)}(x+1)$ and the second formula follows. □

The formula is striking: the mth derivative is not an approximation or an asymptotic rescaling of the original function; it is an exact rescaling of the signed extension. The large factor $2^{m(m+1)/2}$ quantifies the rapid growth of high derivatives, while the argument $2^m x$ accounts for the increasingly fine oscillatory structure.

6.3 Polynomial behavior near dyadic points

Let $x_0 = a/2^n$ be a dyadic point in the interior of the unit interval. For $m > n$, the argument $2^m x_0$ in (50) is an even integer. The local translate formula (12) and flatness of up at its endpoints imply

$$F^{(m)}(x_0) = 0 \qquad (m > n). \tag{52}$$

Thus the Taylor series of F at a dyadic point terminates.

Proposition 6.3 (Terminating Taylor series)**.** *If $x_0 = a/2^n \in (0, 1)$ with a odd, then the formal Taylor series of F at x_0 is a polynomial of degree at most n. Nevertheless F does not agree with that polynomial on any neighborhood of x_0.*

Proof. Equation (52) proves termination. If F agreed with the Taylor polynomial on a nonempty interval, analyticity of the polynomial and the differential scaling relation would propagate polynomial behavior across successively enlarged dyadic intervals. Ultimately F would be polynomial on $(0, 1)$, which is incompatible with its flatness at 0 and its nonconstant value $F(1/2) = 1/2$. Equivalently, a polynomial with all positive-order derivatives zero at 0 is constant, whereas F is not. ☐

Corollary 6.4 (Nowhere analytic)**.** *Neither F on $(0, 1)$ nor* up *on $(-1, 1)$ is real analytic at any point.*

Proof. Dyadic points are dense. If F were analytic at one point, the differential equations and translation/reflection identities would extend analyticity to an interval and then to dyadic points in that interval, contradicting the preceding proposition. The statement for up follows from the fold relation. ☐

7 Moments and generating functions

Exact arithmetic enters through two rational moment sequences. One is attached to the even moments of up; the other records its one-sided moments and, at the same time, the values of F at reciprocal powers of two.

7.1 The even moments c_n

Recall

$$c_n = \int_{-1}^{1} x^{2n}\,\mathrm{up}(x)\,\mathrm{d}x, \qquad c_0 = 1.$$

Theorem 7.1 (Moment recurrence)**.** *For $n \geq 1$,*

$$(2n + 1)(2^{2n} - 1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k. \tag{53}$$

Consequently every c_n is rational and positive.

Proof. Insert the power series (46) into (39). Since

$$\mathrm{sinc}(\pi z) = \sum_{j \geq 0} \frac{(-1)^j (\pi z)^{2j}}{(2j + 1)!}$$

and

$$\widehat{\mathrm{up}}(z/2) = \sum_{k \geq 0} \frac{(-1)^k c_k (\pi z)^{2k}}{(2k)!},$$

comparison of the coefficient of $(-1)^n(\pi z)^{2n}$ gives

$$\frac{2^{2n}c_n}{(2n)!} = \sum_{k=0}^{n} \frac{c_k}{(2k)!(2n-2k+1)!}.$$

Multiplying by $(2n+1)!$ yields

$$(2n+1)2^{2n}c_n = \sum_{k=0}^{n} \binom{2n+1}{2k} c_k.$$

The term $k = n$ is $(2n+1)c_n$; moving it to the left gives (53). Positivity follows either from the integral definition or by induction from the recurrence. $\square$

The first values are

$$1, \quad \frac{1}{9}, \quad \frac{19}{675}, \quad \frac{583}{59535}, \quad \frac{132809}{32531625}, \ldots \tag{54}$$

A useful denominator clearing is

$$c_n = \frac{C_n}{(2n+1)!! \prod_{j=1}^{n}(2^{2j}-1)}, \tag{55}$$

where C_n is a positive integer. It can be defined without division by

$$C_0 = 1, \tag{56}$$

$$C_{n+1} = \sum_{k=0}^{n} C_k \binom{2n+3}{2k} \prod_{j=k+1}^{n}(2j+1) \prod_{j=k+1}^{n}(2^{2j}-1). \tag{57}$$

The formal development proves directly that every C_n is odd. Modulo 2, all factors in the two products are 1, and Lucas's theorem shows that among the coefficients $\binom{2n+3}{2k}$ with $0 \le k \le n$ an odd number are odd. This closes the induction.

7.2 The half moments d_n

Define

$$d_0 = 1, \qquad d_n = n\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t \quad (n \ge 1). \tag{58}$$

Introduce the entire exponential generating function

$$G(z) = 1 + z\int_0^1 \mathrm{up}(t)\mathrm{e}^{zt}\,\mathrm{d}t = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!}. \tag{59}$$

Theorem 7.2 (Generating-function equation)**.** *For every $z \in \mathbb{C}$,*

$$G(2z) = \frac{\mathrm{e}^z - 1}{z}\,G(z), \tag{60}$$

where the quotient has its removable value 1 at $z = 0$.

Proof. One may derive the identity directly from (21) by integration by parts. An especially transparent route uses Fourier analysis. From evenness and the fold relation,

$$G(z) = \mathrm{e}^{z/2}\widehat{\mathrm{up}}\left(\frac{\mathrm{i}z}{4\pi}\right). \tag{61}$$

Applying (39) at $\mathrm{i}z/(2\pi)$ and simplifying $\operatorname{sinc}(\mathrm{i}z/2) = 2\sinh(z/2)/z$ gives

$$G(2z) = \mathrm{e}^z \frac{2\sinh(z/2)}{z}\widehat{\mathrm{up}}\left(\frac{\mathrm{i}z}{4\pi}\right) = \frac{\mathrm{e}^z-1}{z}G(z).$$

$\square$

Corollary 7.3 (Half-moment recurrence). *For* $n \geq 1$,

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \tag{62}$$

In particular $d_n \in \mathbb{Q}$.

Proof. Expand $(\mathrm{e}^z - 1)/z = \sum_{j \geq 0} z^j/(j+1)!$ in (60). Comparison of the coefficient of $z^n/n!$ gives

$$2^n d_n = \frac{1}{n+1} \sum_{k=0}^{n} \binom{n+1}{k} d_k.$$

Isolating the $k = n$ term proves the recurrence. □

The first values are

$$1, \quad \frac{1}{2}, \quad \frac{5}{18}, \quad \frac{1}{6}, \quad \frac{143}{1350}, \quad \frac{19}{270}, \quad \frac{1153}{23814}, \ldots \tag{63}$$

As for c_n, one may clear denominators without division:

$$d_n = \frac{D_n}{(n+1)! \prod_{j=1}^{n} (2^j - 1)}, \tag{64}$$

where $D_n \in \mathbb{N}$. A division-free recurrence follows by substituting (64) into (62). The normalized parity theorem says that, for $n \geq 1$, the reduced numerator and denominator of $2d_n$ are both odd.

7.3 Relation between the two moment sequences

Expanding (61) in powers of z gives a finite binomial transform at each order.

Proposition 7.4. *For every* $n \geq 0$,

$$d_n = 2^{-n} \sum_{0 \leq k \leq n/2} \binom{n}{2k} c_k. \tag{65}$$

Proof. By (46),

$$\widehat{\mathrm{up}}\left(\frac{\mathrm{i}z}{4\pi}\right) = \sum_{k \geq 0} c_k \frac{(z/2)^{2k}}{(2k)!}.$$

Multiplying by $\mathrm{e}^{z/2}$ and extracting the coefficient of $z^n/n!$ gives exactly (65). □

7.4 Bernoulli-number recurrences

Taking logarithms in (60) and using

$$\log \frac{\mathrm{e}^z - 1}{z} = \frac{z}{2} + \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} z^{2m} \tag{66}$$

produces recurrences in which Bernoulli numbers replace binomial convolutions. One convenient form for the even moments is obtained by taking the logarithmic derivative of the sinc product. If

$$\widehat{\mathrm{up}}(z) = \sum_{n \geq 0} (-1)^n c_n \frac{(2\pi z)^{2n}}{(2n)!},$$

then the coefficients satisfy a triangular recurrence whose mth logarithmic coefficient is proportional to

$$\frac{B_{2m}}{m(2m)!} \frac{1}{1 - 2^{-2m}}. \tag{67}$$

The formal development proves both the ordinary convolution recurrences and these Bernoulli forms, and proves their equivalence by formal power-series algebra. The binomial recurrences are usually better for exact computation; the Bernoulli forms are often better for asymptotic estimates.

8 Exact values at reciprocal powers of two

The half moments are already endpoint values in disguise.

Theorem 8.1 (Reciprocal-power values). *For every* $n \geq 0$,

$$F(2^{-n}) = \frac{d_n}{n!\,2^{\binom{n}{2}}}. \tag{68}$$

Equivalently,

$$d_n = n!\,2^{\binom{n}{2}} F(2^{-n}). \tag{69}$$

Proof. For $n \geq 1$, repeated integration by parts using a chain of primitives generated by (21) gives

$$\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t = (n-1)!\,2^{\binom{n}{2}} F(2^{-n}). \tag{70}$$

Multiplying by n and using (58) proves the formula. The case $n = 0$ is $F(1) = d_0 = 1$. □

The first values are therefore

n	$F(2^{-n})$
0	1
1	$\frac{1}{2}$
2	$\frac{5}{72}$
3	$\frac{1}{288}$
4	$\frac{143}{2073600}$
5	$\frac{19}{33177600}$
6	$\frac{1153}{561842749440}$
7	$\frac{583}{179789679820800}$

(71)

The denominators grow far faster than exponentially, in agreement with the quadratic leading term in the logarithmic asymptotic derived later.

8.1 An exact two-adic valuation

The parity theorem for d_n immediately determines the power of two in the reduced denominator.

Theorem 8.2 (Exact 2-adic valuation). *For* $n \geq 1$,

$$v_2\big(F(2^{-n})\big) = -\binom{n}{2} - 1 - v_2(n!). \tag{72}$$

Proof. The reduced numerator and denominator of $2d_n$ are odd, so $v_2(d_n) = -1$. Apply (68) and add valuations:

$$v_2(F(2^{-n})) = v_2(d_n) - v_2(n!) - \binom{n}{2}.$$

□

Legendre's formula

$$v_2(n!) = n - \mathrm{wt}_2(n) \tag{73}$$

then makes the binary dependence explicit.

8.2 A normalized odd integer

For $n \geq 1$ define

$$R_n = 2^{\binom{n-1}{2}} (2n)! F(2^{-n}) \prod_{j=1}^{\lfloor n/2 \rfloor} (2^{2j} - 1). \tag{74}$$

The exponent of two in (74) is *positive*; a negative sign that appears in an abstract of the source paper is a typographical error.

Theorem 8.3. *Each R_n is an odd positive integer. Moreover it has a finite expression in the moment numerators C_k and the odd Mersenne products.*

Proof architecture. Insert (65) into (68), then substitute the normalized form (55). After factorial and double-factorial cancellations, every term has an integral common denominator that is killed by the product in (74). Modulo 2, all Mersenne and odd-factor products are 1, and Lucas's theorem reduces the sum to an odd count of odd binomial coefficients. Thus the resulting integer is odd. □

9 Exact evaluation at arbitrary dyadic rationals

The rationality theorem for dyadic arguments is much stronger than the statement "F happens to take rational values there." It comes with two complementary exact algorithms:

1. a closed finite formula involving Thue–Morse signs and the rational moments c_k;
2. a bit-recursive Taylor algorithm whose number of recursive calls is the number of set bits in the numerator.

The first is best for proofs; the second is best for computation.

9.1 The closed Thue–Morse formula

Theorem 9.1 (Exact dyadic formula)**.** *Let $n, a \in \mathbb{N}$ with $0 \leq a \leq 2^n$. Then*

$$\boxed{F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} (-1)^{\mathrm{wt}_2(h)} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a - 2h - 1)^{n-2k}.} \tag{75}$$

An empty outer sum has value 0.

Proof. Write $x = a/2^n$ and use $F(x) = \mathrm{up}(x-1)$. Since up and all its derivatives vanish at -1, Taylor's theorem with integral remainder gives

$$\mathrm{up}(x-1) = \frac{1}{n!} \int_{-1}^{x-1} (x - 1 - t)^n \, \mathrm{up}^{(n+1)}(t) \, \mathrm{d}t. \tag{76}$$

Insert the exact derivative identity (51):

$$\mathrm{up}^{(n+1)}(t) = 2^{\binom{n+2}{2}} \mathcal{F}\big(2^{n+1}(t+1)\big).$$

Put $s = 2^{n+1}(t+1)$. As t runs from -1 to $x-1$, s runs from 0 to $2a$. Split this interval into the blocks $[2h, 2h+2]$, $0 \leq h < a$. On the hth block, (12) gives

$$\mathcal{F}(s) = (-1)^{\mathrm{wt}_2(h)} \, \mathrm{up}(s - 2h - 1).$$

The substitution $u = s - 2h - 1$ maps the block to $[-1, 1]$. Gathering the powers of two gives

$$F(a/2^n) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} (-1)^{\mathrm{wt}_2(h)} \int_{-1}^{1} (2a - 2h - 1 - u)^n \,\mathrm{up}(u)\,\mathrm{d}u. \tag{77}$$

Expand the nth power. Odd powers of u integrate to zero because up is even, while

$$\int_{-1}^{1} u^{2k} \,\mathrm{up}(u)\,\mathrm{d}u = c_k.$$

The surviving even terms are exactly the inner sum in (75). □

Every quantity on the right of (75) is rational, so dyadic rationality is immediate. More importantly, the formula lives entirely in $\mathbb{Q}$ and can be evaluated without roundoff.

9.2 Thue–Morse cancellation

The mechanism that makes the dyadic formulas simplify is the Prouhet cancellation carried by the Thue–Morse signs.

Lemma 9.2 (Affine power cancellation)**.** *If $d < b$, then for all $X, Y \in \mathbb{Q}$,*

$$\sum_{h=0}^{2^b-1} (-1)^{\mathrm{wt}_2(h)} (X + Yh)^d = 0. \tag{78}$$

Proof. It is enough to prove the claim for h^j, $0 \le j \le d$, and then expand the affine power. Let

$$S_{b,j} = \sum_{h=0}^{2^b-1} (-1)^{\mathrm{wt}_2(h)} h^j.$$

Split the sum for $S_{b+1,j}$ at 2^b and use $\varepsilon_{2^b+h} = -\varepsilon_h$:

$$\begin{aligned} S_{b+1,j} &= \sum_{h<2^b} \varepsilon_h \big(h^j - (2^b + h)^j\big) \\ &= -\sum_{r=0}^{j-1} \binom{j}{r} 2^{b(j-r)} S_{b,r}. \end{aligned}$$

If $j < b + 1$, every index r on the right satisfies $r < b$, so the induction hypothesis makes all terms zero. The base case is immediate. □

A compact generating-function version is

$$\sum_{h<2^b} (-1)^{\mathrm{wt}_2(h)} \mathrm{e}^{ht} = \prod_{j=0}^{b-1} (1 - \mathrm{e}^{2^j t}), \tag{79}$$

whose right side has a zero of order b at $t = 0$. Differentiating fewer than b times at zero gives the same cancellation.

This lemma explains why apparently large translated sums in (75) collapse to short Taylor blocks when the numerator crosses a binary boundary.

9.3 A Taylor block that removes the leading bit

Suppose $0 < x < 1$, and let r be the unique integer such that

$$2^{-r} \leq x < 2^{-r+1}. \tag{80}$$

Put $y = x - 2^{-r}$. Then $0 \leq y < 2^{-r}$.

Theorem 9.3 (Exact Taylor-block recursion)**.** *With the notation above,*

$$F(x) = -F(y) + \sum_{j=0}^{r} \frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} y^j. \tag{81}$$

Proof. Apply Taylor's theorem at the point 2^{-r} through degree r:

$$F(2^{-r} + y) = \sum_{j=0}^{r} \frac{F^{(j)}(2^{-r})}{j!} y^j + R_r(y). \tag{82}$$

The derivative scaling formula gives

$$F^{(j)}(2^{-r}) = 2^{\binom{j+1}{2}} F(2^{j-r}), \qquad 0 \leq j \leq r. \tag{83}$$

It remains to identify the integral remainder. The scaling equation carries the interval $[2^{-r}, 2^{-r} + y]$ into the adjacent signed translate of the interval $[0, y]$; on that translate the Thue–Morse sign is negative. Reversing the substitutions in the derivative identity shows

$$R_r(y) = -\frac{1}{r!} \int_0^y (y - t)^r F^{(r+1)}(t) \, \mathrm{d}t = -F(y),$$

where the last equality is Taylor's integral formula at the flat point 0. Substitution in (82) proves (81). ☐

Using (68), the coefficients may be written entirely in terms of the half moments:

$$\frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} = \frac{2^{\binom{j+1}{2} - \binom{r-j}{2}} d_{r-j}}{j!(r-j)!}. \tag{84}$$

This is the form that appears in the original arithmetic recurrence.

9.4 The bit-recursive evaluator

Let $x = a/2^e$ with $0 < a < 2^e$. Write

$$b = \lfloor \log_2 a \rfloor, \qquad r = e - b, \qquad q = a - 2^b. \tag{85}$$

Then 2^{-r} is the leading binary term of x and

$$y = \frac{q}{2^e}. \tag{86}$$

Applying (81) reduces $F(a/2^e)$ to $F(q/2^e)$; the highest set bit of the numerator has disappeared.

Algorithm 9.4 (Exact dyadic evaluator). Precompute $V_j = F(2^{-j})$ for $0 \le j \le e$ from (68). Define $\mathrm{Eval}(e, a)$ recursively by

$$\mathrm{Eval}(e,a) = \begin{cases} 0, & a \le 0, \\ 1, & a \ge 2^e, \\ T_r(q/2^e) - \mathrm{Eval}(e,q), & 0 < a < 2^e, \end{cases}$$

where b, r, q are given by (85) and

$$T_r(z) = \sum_{j=0}^{r} \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} z^j. \tag{87}$$

Evaluate T_r by Horner's rule.

A literal pseudocode version is:

```
Eval(e, a, V):
    if a <= 0:      return 0
    if a >= 2^e:    return 1

    b = floor_log2(a)
    r = e - b
    q = a - 2^b
    y = q / 2^e

    p = V[0]                         // highest Taylor coefficient, j = r
    for m = 1 .. r:
        j = r - m
        p = V[m] + (2^(j+1) * y / (j+1)) * p

    return p - Eval(e, q, V)
```

The displayed loop is one convenient Horner ordering; any exact evaluation of (87) is equivalent.

Theorem 9.5 (Correctness and termination). *For every $e \in \mathbb{N}$ and $a \in \mathbb{Z}$,*

$$\mathrm{Eval}(e,a) = F(a/2^e) \tag{88}$$

under the bounded convention $F = 0$ to the left and $F = 1$ to the right. The recursion makes exactly $\mathrm{wt}_2(a)$ nontrivial calls when $0 \le a < 2^e$.

Proof. The boundary clauses agree with (1) and (2). In the interior, (81) is precisely the recursive clause. Since $0 \le q < a$, strong induction proves correctness and termination. Replacing a by q clears its highest set bit and leaves all lower bits unchanged, so the number of nontrivial steps is the popcount $\mathrm{wt}_2(a)$. □

Precomputing the reciprocal-power table costs $O(e^2)$ rational operations using the triangular recurrence for d_n. A single value then costs $O(e\ \mathrm{wt}_2(a))$ arithmetic operations in a straightforward implementation. This is much smaller than the a outer summands in (75) when a is large.

Example 9.6 (The value at $5/16$). Here $a = 5 = 2^2 + 1$, $e = 4$, so $r = 2$, $q = 1$, and $y = 1/16$. Formula (81) gives

$$\begin{aligned} F(5/16) &= F(1/4) + 2F(1/2)y + \frac{2^3 F(1)}{2} y^2 - F(1/16) \\ &= \frac{5}{72} + \frac{1}{16} + \frac{1}{64} - \frac{143}{2073600} \\ &= \boxed{\frac{305857}{2073600}}. \end{aligned}$$

Only two recursive calls are needed, corresponding to the two set bits of 5.

9.5 A complete level-4 table

The exact values at denominator 16 are shown below. The second half follows from $F(1-x) = 1 - F(x)$.

x	$F(x)$	x	$F(x)$
0	0	$1/2$	$1/2$
$1/16$	$143/2073600$	$9/16$	$1295857/2073600$
$1/8$	$1/288$	$5/8$	$215/288$
$3/16$	$46657/2073600$	$11/16$	$1767743/2073600$
$1/4$	$5/72$	$3/4$	$67/72$
$5/16$	$305857/2073600$	$13/16$	$2026943/2073600$
$3/8$	$73/288$	$7/8$	$287/288$
$7/16$	$777743/2073600$	$15/16$	$2073457/2073600$

9.6 Denominator bounds

Let

$$M_m = \prod_{j=1}^{m}(2^{2j}-1), \qquad (2m+1)!! = 1 \cdot 3 \cdots (2m+1). \tag{89}$$

Theorem 9.7 (Uniform denominator multiplier). *Let $n \geq 1$ and $a \in \mathbb{Z}$ with $|a| < 2^n$. Then*

$$\mathrm{up}(a/2^n)\; n!\, 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!!\, M_{\lfloor n/2 \rfloor} \tag{90}$$

is an integer.

Proof. Apply the global dyadic form of (75). Replace each c_k by (55). For $k \leq \lfloor n/2 \rfloor$, both the odd double factorial in the denominator of c_k and its even-Mersenne product divide the two common products in (90). The remaining factors are integers. Thus every summand becomes integral after multiplication by the displayed common factor. □

Let Δ_n be the least common multiple of the reduced denominators of the values $\mathrm{up}(a/2^n)$ with odd a in the support. The theorem gives the explicit divisibility bound

$$\Delta_n \mid n!\, 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!! M_{\lfloor n/2 \rfloor}. \tag{91}$$

The exact closed form of Δ_n is subtler. The source paper records a conjectural factorization; the formal development correctly retains it as a conjecture rather than a theorem.

9.7 A q-binomial form and translation invariance

For completeness, the formal development also derives a less obvious formula organized by Gaussian binomial coefficients. Put

$$(a;q)_n = \prod_{j=0}^{n-1}(1 - aq^j), \tag{92}$$

and write $\begin{bmatrix} n \\ k \end{bmatrix}_q$ for the q-binomial coefficient.

Theorem 9.8 (q-binomial dyadic formula). *For $m, n \in \mathbb{N}$ and every scalar translation $Q \in \mathbb{Q}$,*

$$F(m/2^n) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{m2^k-1} (-1)^{\mathrm{wt}_2(r)} \left(r - m2^k + Q\right)^{n+k}, \tag{93}$$

whenever $0 \leq m \leq 2^n$. The right side is independent of Q.

Proof architecture. Repeated use of the moment recurrence converts the moment kernel in (75) into the Gaussian-binomial coefficients in (93). To prove translation independence, regard the right side as a polynomial in Q. Its positive-order derivatives are linear combinations of translated Thue–Morse power sums. The cancellation lemma (78), together with the triangular q-binomial identity, makes every derivative vanish. Hence the polynomial is constant. The formal proof packages this polynomial over $\mathbb{Q}$ and then extends the identity by scalar evaluation to any characteristic-zero field, including $\mathbb{R}$ and $\mathbb{C}$. □

The q-binomial formula is not competitive with the bit recursion as an evaluator. Its value is structural: it reveals a hidden finite-difference symmetry and shows that the same exact identity survives scalar extension.

10 Endpoint asymptotics: exact logarithmic decomposition

We now turn to the behavior of $F(x)$ as $x \downarrow 0$. This is the technically deepest part of the theory and the place where a very small, but mathematically unavoidable, log-periodic fluctuation appears.

Throughout the asymptotic sections, put

$$L = \log 2. \tag{94}$$

We use the standard Stieltjes convention

$$\zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + \mathcal{O}(s^2), \tag{95}$$

where γ is Euler's constant and γ_1 is the first Stieltjes constant.

10.1 The negative Laplace product

The function G from (59) is the moment generating function of the random variable X in (26):

$$G(z) = \mathbb{E}\mathrm{e}^{zX}. \tag{96}$$

Indeed, this also follows from (61) and $Y = 2X - 1$. Therefore, for $s > 0$, $G(-s)$ is the ordinary Laplace transform of the probability law.

Iterating the functional equation (60) in the negative direction gives an exact product.

Proposition 10.1 (Negative Laplace product). *For $s > 0$,*

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - \mathrm{e}^{-s/2^j}}{s/2^j}. \tag{97}$$

Consequently

$$\Lambda(s) := \log G(-s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \qquad \kappa(u) = \log \frac{1 - \mathrm{e}^{-u}}{u}. \tag{98}$$

The series converges absolutely and locally uniformly on $(0, \infty)$.

Proof. Putting $z = -s/2$ in (60) gives

$$G(-s) = \frac{1 - \mathrm{e}^{-s/2}}{s/2} G(-s/2).$$

After N iterations, the remaining factor is $G(-s/2^N)$, which tends to $G(0) = 1$. For small u,

$$\kappa(u) = -\frac{u}{2} + \mathcal{O}(u^2),$$

so the logarithmic series is dominated on compact s-intervals by a geometric series. □

The product has the exact dilation law

$$\Lambda(2s) = \kappa(s) + \Lambda(s). \tag{99}$$

This equation resembles a renewal equation on the logarithmic scale. Its homogeneous solutions are periodic functions of $\log_2 s$, which is the conceptual origin of the fluctuation in the final asymptotic.

10.2 Removing the quadratic drift

Define the forward tail

$$T(s) = \sum_{m=0}^{\infty} \log(1 - \mathrm{e}^{-s2^m}) < 0 \tag{100}$$

and the positive tail error

$$E(s) = -T(s) \geq 0. \tag{101}$$

The series is exponentially small as $s \to \infty$.

Lemma 10.2 (Forward-tail bound)**.** *For $s > 0$,*

$$0 \leq E(s) \leq \frac{\mathrm{e}^{-s}}{(1 - \mathrm{e}^{-s})^2}. \tag{102}$$

In particular $E(s) \leq 4\mathrm{e}^{-s}$ for $s \geq \log 2$.

Proof. For $0 < q < 1$, $-\log(1 - q) \leq q/(1 - q)$. Hence

$$E(s) \leq \frac{1}{1 - \mathrm{e}^{-s}} \sum_{m \geq 0} \mathrm{e}^{-s2^m}.$$

Since $2^m \geq m + 1$, the last sum is at most $\sum_{m \geq 0} \mathrm{e}^{-s(m+1)} = \mathrm{e}^{-s}/(1 - \mathrm{e}^{-s})$. □

For $t \in \mathbb{R}$, set

$$P(t) = \Lambda(2^t) + \frac{L}{2}(t^2 - t) + T(2^t). \tag{103}$$

Theorem 10.3 (Exact periodic decomposition)**.** *The function P is 1-periodic and smooth. For every $s > 0$, with $t = \log_2 s$, one has the exact identity*

$$\boxed{\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s).} \tag{104}$$

Proof. Write $s = 2^t$. Using (99),

$$\Lambda(2s) = \Lambda(s) + \log(1 - \mathrm{e}^{-s}) - \log s.$$

Also

$$T(2s) = T(s) - \log(1 - \mathrm{e}^{-s}).$$

The change in the quadratic correction is

$$\frac{L}{2}\big((t+1)^2 - (t+1) - (t^2 - t)\big) = Lt = \log s.$$

The three changes cancel, proving $P(t+1) = P(t)$. Rearranging (103) and using $E = -T$ yields (104). Smoothness follows by differentiating locally uniformly convergent series; periodicity then makes every derivative bounded. □

Let

$$\overline{P} = \int_0^1 P(t)\,\mathrm{d}t, \qquad \Psi(t) = P(t) - \overline{P}. \tag{105}$$

Then Ψ is smooth, real-valued, 1-periodic, and has mean zero.

10.3 Fourier coefficients of the periodic correction

The periodic term is not an unspecified bounded nuisance: all of its Fourier coefficients are explicit. Put

$$\chi_k = \frac{2\pi \mathrm{i} k}{L} \qquad (k \in \mathbb{Z}), \tag{106}$$

and use the convention

$$\widehat{\Psi}(k) = \int_0^1 \Psi(t)\mathrm{e}^{-2\pi \mathrm{i} kt}\,\mathrm{d}t. \tag{107}$$

Theorem 10.4 (Gamma–zeta Fourier coefficients)**.** *One has $\widehat{\Psi}(0) = 0$, and for $k \neq 0$,*

$$\boxed{\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}.} \tag{108}$$

Every nonzero Fourier coefficient is nonzero. In particular, Ψ is genuinely nonconstant.

Proof. The zero coefficient is zero by definition. For $k \neq 0$, change variables $s = 2^t$ in (107). Dyadic periodization turns the integral over one logarithmic period into a Mellin transform. The basic Mellin identity is

$$\int_0^\infty s^{z-1} \log(1 - \mathrm{e}^{-s})\,\mathrm{d}s = -\Gamma(z)\zeta(1+z), \qquad \Re z > 0. \tag{109}$$

It follows by expanding $\log(1 - \mathrm{e}^{-s}) = -\sum_{m \geq 1} \mathrm{e}^{-ms}/m$ and integrating term by term. Near $s = 0$ the kernel contains logarithmic and linear divergences; the quadratic correction in (103) is exactly the finite-part subtraction that removes them. Analytic continuation of the regularized Mellin transform to the imaginary axis gives (108) at $z = -\chi_k$.

The Gamma function has no zeros. Moreover $1 - \chi_k$ lies on the line $\Re z = 1$ away from the pole at 1, and the zeta function has no zeros there. Hence the product in (108) is nonzero. □

Conjugate symmetry $\widehat{\Psi}(-k) = \overline{\widehat{\Psi}(k)}$ reconstructs a real-valued Fourier series. Stirling's estimate for Γ on vertical lines implies exponential decay of the coefficients, so the Fourier series and all of its differentiated series converge rapidly.

The first coefficient is tiny:

$$\widehat{\Psi}(1) \approx (7.5586467408 - 7.4700963646\,\mathrm{i}) \times 10^{-7}, \tag{110}$$

with magnitude approximately $1.0627110626 \times 10^{-6}$. Thus the visible oscillation is on the scale of a few parts in a million in the logarithm. Tiny is not the same as zero: its nonvanishing changes the logical status of an asymptotic equivalent.

10.4 The finite part at the Mellin origin

The constant that enters the saddle asymptotic comes from the finite part of $-\Gamma(z)\zeta(1+z)$ at $z = 0$. Using

$$\begin{aligned}\Gamma(z) &= \frac{1}{z} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12}\right) z + \mathcal{O}(z^2), \\ \zeta(1+z) &= \frac{1}{z} + \gamma - \gamma_1 z + \mathcal{O}(z^2),\end{aligned}$$

one finds

$$-\Gamma(z)\zeta(1+z) + \frac{1}{z^2} = A_0 + \mathcal{O}(z), \tag{111}$$

where

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}. \tag{112}$$

The final constant is

$$C_F = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log\pi \approx -2.02798389863885942397. \tag{113}$$

The terms $-7L/12$ and $-(1/2)\log\pi$ arise from the dyadic Euler–Maclaurin finite part and the leading Gaussian saddle mass, respectively.

11 The corrected sharp asymptotic

11.1 The lower Lambert coordinate

For sufficiently small $x > 0$, define

$$W = W_{-1}(-Lx), \tag{114}$$

where W_{-1} is the lower real branch of Lambert's W-function, and put

$$\lambda = \lambda(x) = -\frac{W}{L}. \tag{115}$$

Then $\lambda \to \infty$ as $x \downarrow 0$ and

$$\lambda 2^{-\lambda} = x. \tag{116}$$

Equivalently,

$$r = 2^{\lambda} = \frac{\lambda}{x}. \tag{117}$$

The real number r is the explicit Lambert saddle radius used in the contour analysis.

11.2 The main term

Define

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda), \tag{118}$$

where C_F is given by (113) and $\lambda = \lambda(x)$. The same expression can be written compactly in the Lambert variable W:

$$\mathcal{M}(x) = -\frac{7L}{12} - \frac{1}{2}\log(\pi x) + \frac{A_0 - W - \frac{1}{2}W^2}{L} + \Psi\left(-\frac{W}{L}\right). \tag{119}$$

The equivalence follows from $W = -L\lambda$ and $\log x = \log\lambda - L\lambda$.

Theorem 11.1 (Corrected sharp endpoint asymptotic)**.** *As $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{\lambda(x)}\right) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{-\log x}\right).} \tag{120}$$

Equivalently,

$$F(x) \sim \exp\mathcal{M}(x). \tag{121}$$

The periodic term is evaluated at the *exact* phase $\lambda(x)$, not at a truncated log–log approximation to it. This matters at the stated error scale.

11.3 Bromwich inversion

Because $G(-z) = \mathbb{E}e^{-zX}$ is entire and F is continuous, Laplace–Stieltjes inversion gives, for every $r > 0$ and $0 < x < 1$,

$$F(x) = \frac{1}{2\pi i}\int_{r-i\infty}^{r+i\infty} \frac{e^{zx}G(-z)}{z}\,dz. \tag{122}$$

Set $z = r(1 + i\theta)$ and choose $r = 2^\lambda = \lambda/x$. Then

$$F(x) = \frac{e^{\lambda+\Lambda(r)}}{2\pi}\int_{-\infty}^{\infty} e^{i\lambda\theta}\frac{G(-r(1+i\theta))}{G(-r)}\frac{d\theta}{1+i\theta}. \tag{123}$$

The exact real decomposition (104) at $r = 2^\lambda$ gives

$$\lambda + \Lambda(r) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda + P(\lambda) + E(r). \tag{124}$$

Since

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \qquad \overline{P} - \frac{1}{2}\log(2\pi) = C_F, \tag{125}$$

the missing factors in (118) are exactly the leading Gaussian mass $1/\sqrt{2\pi\lambda}$ and the centered fluctuation $\Psi = P - \overline{P}$.

11.4 The local phase

Let

$$u = \log(1 + i\theta), \tag{126}$$

with the principal logarithm near the saddle. Subtracting the value at $\theta = 0$ from the complexified version of (104), and including the factor $(1 + i\theta)^{-1}$ in (123), gives the local phase

$$\Phi_\lambda(\theta) = \lambda(i\theta - u) - \frac{u}{2} - \frac{u^2}{2L} + \Psi\left(\lambda + \frac{u}{L}\right) - \Psi(\lambda) + \mathcal{E}_r(\theta), \tag{127}$$

where $\mathcal{E}_r(0) = 0$ and, on a fixed neighborhood of zero, every derivative of $\mathcal{E}_r$ is exponentially small in r. This last term comes from the forward tail E and is far smaller than every power of $1/\lambda$.

The elementary expansion

$$\log(1 + \mathrm{i}\theta) = \mathrm{i}\theta + \frac{\theta^2}{2} - \frac{\mathrm{i}\theta^3}{3} - \frac{\theta^4}{4} + \mathcal{O}(\theta^5) \tag{128}$$

shows that

$$\lambda(\mathrm{i}\theta - u) = -\frac{\lambda\theta^2}{2} + \frac{\mathrm{i}\lambda\theta^3}{3} + \frac{\lambda\theta^4}{4} + \mathcal{O}(\lambda|\theta|^5). \tag{129}$$

Thus the natural central scale is

$$\theta = \frac{v}{\sqrt{\lambda}}. \tag{130}$$

Uniformly for $|v|$ growing sufficiently slowly with λ,

$$\Phi_\lambda(v/\sqrt{\lambda}) = -\frac{v^2}{2} + \lambda^{-1/2} P_1(v;\lambda) + \lambda^{-1} P_2(v;\lambda) + \mathcal{O}\Big(\lambda^{-3/2}(1 + |v|^9)\Big), \tag{131}$$

where

$$A(u) = \frac{\Psi'(u)}{L} - \frac{1}{2}, \tag{132}$$

$$B(u) = -\frac{1}{4} + \frac{1}{2L} + \frac{\Psi'(u)}{2L} - \frac{\Psi''(u)}{2L^2}, \tag{133}$$

$$P_1(v;u) = \mathrm{i}\left(A(u)v + \frac{v^3}{3}\right), \tag{134}$$

$$P_2(v;u) = \frac{v^4}{4} + B(u)v^2. \tag{135}$$

In (131), the second argument is $u = \lambda$.

11.5 Central and tail estimates

For completeness, the quantitative saddle proof is organized by the following estimates. They are stated here in the form needed for the argument; the formal development proves them with explicit domains and constants.

Lemma 11.2 (Central expansion). *For every fixed truncation order N, there is $\delta > 0$ such that on $|\theta| \leq \delta$ the exponent in (123) admits a Taylor expansion through order N after the scaling (130). Its coefficients are smooth 1-periodic functions of λ, and the remainder is dominated by a polynomial in v times the next power of $\lambda^{-1/2}$.*

Proof. The logarithm in (126) is analytic for $|\theta| < 1$. The periodic function Ψ is smooth with bounded derivatives of every order. Taylor's theorem applied to each term of (127), together with the exponentially small bounds on $\mathcal{E}_r$, gives a uniform expansion. The dependence on λ occurs only through $\Psi^{(j)}(\lambda)$, hence is periodic. □

Lemma 11.3 (Gaussian domination). *There are positive constants c, δ and λ_0 such that, for $\lambda \geq \lambda_0$ and $|\theta| \leq \delta$,*

$$\Re\Phi_\lambda(\theta) \leq -c\lambda\theta^2. \tag{136}$$

Proof architecture. The exact function $\Re(\mathrm{i}\theta - \log(1+\mathrm{i}\theta)) = -(1/2)\log(1+\theta^2)$ supplies the negative quadratic term. All remaining terms in (127) have bounded first and second derivatives on a fixed neighborhood. For large λ the $-\lambda\theta^2/2$ contribution dominates them. □

Lemma 11.4 (Off-saddle decay)**.** *For each N one may choose the central window so that the contribution to* (123) *outside that window is $O(\lambda^{-N})$ relative to the leading saddle mass.*

Proof architecture. Split the negative-Laplace product at the dyadic index nearest λ. In the intermediate range, a positive proportion of the factors have arguments of order one and their moduli are strictly smaller off the real axis, yielding exponential decay in $\lambda\theta^2$. In the far range, repeated use of the sinc/Laplace product bounds gives arbitrarily high polynomial decay. The two bounds overlap and cover the whole complement of the central window. The factor $(1+\mathrm{i}\theta)^{-1}$ is harmless. □

These lemmas justify extending the scaled central integral to the entire real v-axis and integrating its expansion term by term against $\mathrm{e}^{-v^2/2}$.

11.6 Completion of the leading proof

Keeping only the Gaussian term in (131) gives

$$\int_{\mathbb{R}} \mathrm{e}^{\Phi_\lambda(\theta)}\,\mathrm{d}\theta = \frac{1}{\sqrt{\lambda}}\int_{\mathbb{R}} \mathrm{e}^{-v^2/2}\,\mathrm{d}v\,\big(1+\mathcal{O}(\lambda^{-1})\big) = \sqrt{\frac{2\pi}{\lambda}}\,\big(1+\mathcal{O}(\lambda^{-1})\big)\,. \tag{137}$$

The term of order $\lambda^{-1/2}$ integrates to zero because P_1 is odd in v and purely imaginary. Combining (124), (125), and (137) yields

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1+\frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) + \mathcal{O}(\lambda^{-1}),$$

which is (120). Finally, $\lambda \asymp -\log x$ follows from (116), so the two displayed error scales are equivalent.

12 First correction and the all-orders saddle expansion

The leading theorem leaves an error of order $1/\lambda$. The same contour analysis can be continued to arbitrary order. The coefficient functions are periodic because every local coefficient is built from derivatives of Ψ evaluated at the exact phase λ.

12.1 The first correction

Expand the exponential in (131):

$$\mathrm{e}^{\Phi_\lambda(v/\sqrt{\lambda})} = \mathrm{e}^{-v^2/2}\Big[1 + \lambda^{-1/2}P_1(v;\lambda) \tag{138}$$

$$+\,\lambda^{-1}\big(P_2(v;\lambda) + \tfrac{1}{2}P_1(v;\lambda)^2\big) + \mathcal{O}(\lambda^{-3/2}(1+|v|^{12}))\Big]. \tag{139}$$

Let Z be a standard normal random variable. Since

$$\mathbb{E}Z^2 = 1, \qquad \mathbb{E}Z^4 = 3, \qquad \mathbb{E}Z^6 = 15, \tag{140}$$

the normalized coefficient of λ^{-1} in the saddle mass is

$$\begin{aligned} b_1(u) &= \mathbb{E}\left[P_2(Z;u) + \tfrac{1}{2}P_1(Z;u)^2\right] \\ &= \frac{3}{4} + B(u) - \frac{1}{2}\left(A(u)^2 + 2A(u) + \frac{5}{3}\right). \end{aligned} \tag{141}$$

Substituting (132) and (133) simplifies the expression considerably.

Theorem 12.1 (First saddle correction). *Define*

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \tag{142}$$

Then

$$\boxed{\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda(x))}{\lambda(x)} + \mathcal{O}\big(\lambda(x)^{-2}\big)\,.} \tag{143}$$

Proof. The order-$\lambda^{-1/2}$ term in (139) is odd and integrates to zero. Equation (141) gives the first nonzero relative correction to the integral. A direct simplification yields (142). If the normalized saddle mass is $1 + b_1/\lambda + O(\lambda^{-2})$, then its logarithm is $b_1/\lambda + O(\lambda^{-2})$, so the same coefficient appears in the logarithmic expansion. The off-saddle contribution is smaller than the stated remainder by the tail estimates of the preceding section. ☐

The constant part $1/24 + 1/(2L)$ is the usual type of Gaussian/curvature correction. The terms involving Ψ' and Ψ'' record the fact that the log-periodic environment is not constant across the saddle width.

12.2 All orders

At order N, Taylor expansion of (127) produces finitely many polynomials in v whose coefficients are differential polynomials in

$$\Psi'(u), \Psi''(u), \ldots, \Psi^{(2N)}(u).$$

After exponentiation, all odd powers of $\lambda^{-1/2}$ integrate to zero by parity. The remaining Gaussian moments are integers, so one obtains integer powers of $1/\lambda$.

Theorem 12.2 (Full logarithmic expansion). *There are smooth* 1*-periodic functions* $A_j : \mathbb{R} \to \mathbb{R}$, $j \geq 0$, *with* $A_0 = 0$ *and* A_1 *given by* (142), *such that for every* $N \geq 1$,

$$\boxed{\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}\big(\lambda(x)^{-N}\big)} \tag{144}$$

as $x \downarrow 0$.

Proof architecture. Fix N. Expand the analytic local phase through sufficiently high order in $\lambda^{-1/2}$, with a Gaussian-dominated remainder. Exponentiate using complete Bell polynomials, integrate term by term, and discard odd terms. The quantitative tail theorem makes the complement of the central region $O(\lambda^{-N})$. This gives a relative saddle mass expansion

$$1 + \sum_{j=1}^{N-1} \frac{b_j(\lambda)}{\lambda^j} + O(\lambda^{-N}), \tag{145}$$

where every b_j is smooth and periodic. Taking the formal logarithm defines the A_j. If

$$\log\left(1 + \sum_{j\geq 1} b_j z^j\right) = \sum_{j\geq 1} A_j z^j,$$

then the coefficients may be generated recursively by

$$A_n = b_n - \frac{1}{n}\sum_{k=1}^{n-1} k A_k b_{n-k}. \tag{146}$$

All operations preserve smooth periodic dependence on the phase. □

This form of the expansion is preferable to a very long series in nested logarithms. The exact Lambert coordinate absorbs the dominant implicit inversion, while the exact periodic phase prevents phase errors from being magnified into false remainder claims.

13 Elementary log–log forms and the missing periodic term

Lambert W gives the cleanest formula, but an expansion in elementary logarithms is useful for comparison with published displays. It must, however, retain Ψ at the exact Lambert phase.

13.1 The literal small-x expansion

Put

$$S = -\log x, \qquad m = \log S, \qquad a = \log L. \tag{147}$$

Thus $S \to \infty$ as $x \downarrow 0$. Define the nonperiodic elementary expression

$$\begin{aligned}\mathcal{E}_0(x) = & -\frac{S^2}{2L} - \frac{Sm}{L} + \left(\frac{1}{2} + \frac{1+a}{L}\right) S \\ & - \frac{m^2}{2L} + \frac{am}{L} + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6a^2}{12L} - \frac{7L}{12} - \frac{1}{2}\log\pi \\ & - \frac{m^2}{2LS} + \frac{am}{LS}.\end{aligned} \tag{148}$$

Theorem 13.1 (Corrected elementary expansion)**.** *As $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{E}_0(x) + \Psi(\lambda(x)) + \mathcal{O}\left(\frac{1}{-\log x}\right).} \tag{149}$$

Proof architecture. The standard lower-branch Lambert expansion solves $\lambda L - \log\lambda = S$ by successive substitution. Substituting that expansion into (118), and retaining enough phase terms to control the amplified quadratic contribution, gives (148). A coarse two-term approximation to λ is insufficient: because the main exponent contains $-L\lambda^2/2$, an error $\delta\lambda$ contributes about $-L\lambda\delta\lambda$, so the phase must be known one order more accurately than a naive count suggests. The fourth Lambert phase term in the formal proof is exactly what reduces the residual to $O(1/S)$. □

In the original variable $\ell = \log x < 0$, the same expression is

$$\begin{aligned}\mathcal{E}_0(x) = & -\frac{\ell^2}{2L} + \frac{\ell\log(-\ell)}{L} - \left(\frac{1}{2} + \frac{1+\log L}{L}\right)\ell \\ & - \frac{\log^2(-\ell)}{2L} + \frac{\log L\,\log(-\ell)}{L} \\ & + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6\log^2 L}{12L} - \frac{7L}{12} - \frac{1}{2}\log\pi \\ & + \frac{\log^2(-\ell)}{2L\ell} - \frac{\log L\,\log(-\ell)}{L\ell}.\end{aligned} \tag{150}$$

This is the literal elementary display formalized in the repository, with the periodic term added separately.

13.2 The dyadic logarithmic scale

Set $x = 2^{-t}$ and let

$$\lambda_t = -\frac{1}{L}W_{-1}(-L2^{-t}). \tag{151}$$

Then λ_t is the unique large solution of

$$\lambda_t - \log_2 \lambda_t = t. \tag{152}$$

Its first terms are

$$\lambda_t = t + \frac{\log t}{L} + \frac{\log t}{L^2 t} + \frac{\log t - \frac{1}{2}\log^2 t}{L^3 t^2} + \mathcal{O}\left(\frac{\log^3 t}{t^3}\right). \tag{153}$$

Define

$$\mathcal{D}_0(t) = \; -\frac{L}{2}t^2 - t\log t + \left(1 + \frac{L}{2}\right)t - \frac{\log^2 t}{2L} + C_F - \frac{\log^2 t}{2L^2 t}. \tag{154}$$

Theorem 13.2 (Corrected dyadic-scale expansion)**.** *As $t \to \infty$,*

$$\boxed{\log F(2^{-t}) = \mathcal{D}_0(t) + \Psi(\lambda_t) + \mathcal{O}(t^{-1}).} \tag{155}$$

The literal substitution of $x = 2^{-t}$ into (148) differs from $\mathcal{D}_0(t)$ by the explicit term

$$\frac{\log^2 L}{2L^2 t}, \tag{156}$$

which is itself $O(1/t)$. Thus both nonperiodic displays are valid at the stated precision, but they are not algebraically identical.

13.3 Why the uncorrected formula is false at the claimed scale

Let $\mathcal{M}_0(x) = \mathcal{M}(x) - \Psi(\lambda(x))$ be the nonperiodic Lambert main. From (120),

$$\log F(x) - \mathcal{M}_0(x) = \Psi(\lambda(x)) + O(1/\lambda(x)). \tag{157}$$

Since Ψ is nonconstant, the right side does not tend to zero.

Proposition 13.3 (Failure without the periodic correction)**.** *The relation*

$$\log F(x) = \mathcal{M}_0(x) + O(1/(-\log x)) \tag{158}$$

is false. In fact $F(x)/\exp(\mathcal{M}_0(x))$ has distinct subsequential limits.

Proof. Choose $u, v \in [0, 1)$ with $\Psi(u) \neq \Psi(v)$. Set

$$\lambda_n = n + u, \qquad x_n = \lambda_n 2^{-\lambda_n}.$$

Then $x_n \downarrow 0$ and the exact Lambert phase of x_n is λ_n. Hence

$$\log F(x_n) - \mathcal{M}_0(x_n) \longrightarrow \Psi(u).$$

The analogous sequence with phase $n + v$ tends to $\Psi(v)$. Since $1/(-\log x) \to 0$, the bound (158) is impossible. Exponentiating gives distinct ratio limits $e^{\Psi(u)}$ and $e^{\Psi(v)}$. □

Numerically the discrepancy is minute because the first Fourier mode has size about 10^{-6}, but asymptotic equivalence is a limit statement, not a graphical tolerance. A nonzero bounded oscillation cannot be hidden inside an error term that tends to zero.

13.4 Symmetry at the right endpoint

The symmetry $F(1-x) = 1 - F(x)$ immediately transfers the left-endpoint expansion to the right:

$$1 - F(1-x) = F(x). \tag{159}$$

Thus the same Lambert phase, periodic fluctuation, and all-orders coefficients describe the survival probability near 1.

14 Consequences of the endpoint expansion

The full sharp formula immediately settles several qualitative questions that are otherwise awkward to read from the differential equation.

Theorem 14.1 (Quadratic logarithmic law). *As $x \downarrow 0$,*

$$\frac{\log F(x)}{(\log x)^2} \longrightarrow -\frac{1}{2\log 2}. \tag{160}$$

Equivalently, on the dyadic scale,

$$\log F(2^{-t}) \sim -\frac{\log 2}{2} t^2. \tag{161}$$

Proof. In (118), $\lambda \sim S/L$ with $S = -\log x$. The quadratic term $-L\lambda^2/2$ dominates the linear term, the logarithmic term, and the bounded periodic correction. Substitution gives (160). □

Corollary 14.2 (Between powers and pure endpoint exponentials). *For every $N > 0$ and every $c > 0$,*

$$F(x) = o(x^N), \tag{162}$$

and

$$\mathrm{e}^{-c/x} = o(F(x)) \tag{163}$$

as $x \downarrow 0$.

Proof. The logarithm of x^N is $-NS$, whereas $\log F(x) \sim -S^2/(2L)$, so $F(x)/x^N \to 0$. On the other hand, $\log \mathrm{e}^{-c/x} = -c\mathrm{e}^S$, which is much more negative than $-S^2/(2L)$; therefore $\mathrm{e}^{-c/x}/F(x) \to 0$. □

This places the endpoint decay in an intermediate regime. The function is flatter than any finite algebraic order, as every compactly supported smooth function must be at its support boundary, but it is vastly larger than the familiar model $\mathrm{e}^{-1/x}$.

Corollary 14.3 (Asymptotics of reciprocal-power values). *For integers $n \to \infty$,*

$$\begin{aligned} \log F(2^{-n}) = &-\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ &- \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + O(n^{-1}), \end{aligned} \tag{164}$$

where λ_n is defined by (151). Combining this with (68) gives an equally sharp expansion for the half moments d_n.

The formula shows why a fixed multiplicative constant cannot complete a recurrence-based ansatz for $F(2^{-n})$: the fractional parts of λ_n drift, and the factor $\exp \Psi(\lambda_n)$ is nonconstant.

15 The Fourier–Legendre expansion of up

A second exact spectral representation uses the ordinary Legendre polynomials P_n on $[-1,1]$. Unlike the Fourier transform, this is a polynomial expansion on the natural support interval, and its coefficients can be expressed by the reciprocal-power dyadic values already computed.

15.1 Orthogonality and coefficients

Normalize the Legendre polynomials by $P_n(1)=1$. They satisfy

$$-\frac{\mathrm{d}}{\mathrm{d}x}\big((1-x^2)P_n'(x)\big)=n(n+1)P_n(x), \tag{165}$$

with orthogonality

$$\int_{-1}^{1} P_m(x)P_n(x)\,\mathrm{d}x=\frac{2}{2n+1}\,\delta_{mn}. \tag{166}$$

For a continuous function f, define

$$\alpha_n(f)=\frac{2n+1}{2}\int_{-1}^{1} f(x)P_n(x)\,\mathrm{d}x. \tag{167}$$

Because up is even and $P_n(-x)=(-1)^nP_n(x)$,

$$\alpha_{2n+1}(\mathrm{up})=0. \tag{168}$$

Put

$$u_n=\alpha_{2n}(\mathrm{up})=\frac{4n+1}{2}\int_{-1}^{1}\mathrm{up}(x)P_{2n}(x)\,\mathrm{d}x. \tag{169}$$

15.2 Exact coefficient formula

The even Legendre polynomial has the finite power expansion

$$P_{2n}(x)=4^{-n}\sum_{k=0}^{n}(-1)^{n+k}\binom{2n}{n+k}\binom{2n+2k}{2n}x^{2k}. \tag{170}$$

The moment-to-dyadic bridge is

$$c_k=2(2k)!\,2^{\binom{2k+1}{2}}F(2^{-2k-1}). \tag{171}$$

This is (70) with an even exponent, together with symmetry.

Theorem 15.1 (Exact Legendre coefficients). *For every $n\ge 0$,*

$$u_n=4^{-n}(4n+1)\sum_{k=0}^{n}(-1)^{n+k}\binom{2n}{n+k}\binom{2n+2k}{2n} \times (2k)!\,2^{\binom{2k+1}{2}}F(2^{-2k-1}). \tag{172}$$

In particular every u_n is rational.

Proof. Insert (170) into (169) and interchange the finite sum with the integral. The integral of $x^{2k}\,\mathrm{up}(x)$ is c_k. Substitution of (171) cancels the factor $1/2$ in (169) and gives (172). □

Thus the entire polynomial spectral expansion is anchored in the same exact arithmetic table $F(2^{-m})$ that drives the bit-recursive evaluator.

15.3 Absolute and uniform convergence

The Legendre Sturm–Liouville operator

$$\mathcal{L}f = -\frac{\mathrm{d}}{\mathrm{d}x}\big((1-x^2)f'(x)\big) \tag{173}$$

is formally self-adjoint on $[-1,1]$. Boundary terms vanish because of the factor $1-x^2$. Repeated integration by parts therefore gives, for $n \geq 1$,

$$\alpha_n(f) = \frac{2n+1}{2[n(n+1)]^m}\int_{-1}^{1}(\mathcal{L}^m f)(x)P_n(x)\,\mathrm{d}x. \tag{174}$$

Since $|P_n(x)| \leq 1$ on $[-1,1]$, any C^∞ function has coefficients that decay faster than every power of n.

Theorem 15.2 (Uniform Legendre series)**.** *On the entire closed interval* $[-1,1]$,

$$\boxed{\mathrm{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x).} \tag{175}$$

The series converges absolutely and uniformly, including at both endpoints.

Proof. Apply (174) to $f = \mathrm{up}$. Choosing $m \geq 2$ gives $|\alpha_n(\mathrm{up})| \leq Cn^{-3}$, say. Since $|P_n| \leq 1$, the Weierstrass test gives absolute and uniform convergence. Let the uniform sum be g. Orthogonality shows that g and up have the same Legendre coefficients. Completeness of the Legendre polynomials in $L^2[-1,1]$ implies $g = \mathrm{up}$ almost everywhere, and continuity upgrades equality to every point. Odd coefficients vanish by (168), leaving (175). □

The endpoint statement is worth emphasizing. Generic Fourier–Legendre convergence theorems are often quoted only on the open interval, or with an endpoint average. Here smoothness and absolute uniform convergence give the literal values $\mathrm{up}(\pm 1) = 0$.

15.4 A practical polynomial approximation

Truncating (175) at $n = N$ gives a polynomial of degree $2N$ with exact rational coefficients after expanding the P_{2n}. The error is bounded by

$$\left\| \mathrm{up} - \sum_{n=0}^{N} u_n P_{2n} \right\|_\infty \leq \sum_{n>N} |u_n|. \tag{176}$$

Repeated use of (174) supplies arbitrarily high algebraic tail bounds. For moderate precision this method is attractive: all coefficients are exact, evaluation is stable on $[-1,1]$, and the approximation respects evenness automatically.

15.5 Least-squares optimality of the partial sums

The Legendre truncation is not only a convenient polynomial approximation; it is the exact orthogonal projection in $L^2[-1,1]$. Define

$$S_N(x) = \sum_{n=0}^{N} u_n P_{2n}(x). \tag{177}$$

Theorem 15.3 (Pythagorean error identity). *For every real polynomial p of degree at most $2N+1$,*

$$\int_{-1}^{1} \big(\mathrm{up}(x) - p(x)\big)^2 \,\mathrm{d}x = \int_{-1}^{1} \big(\mathrm{up}(x) - S_N(x)\big)^2 \,\mathrm{d}x + \int_{-1}^{1} \big(p(x) - S_N(x)\big)^2 \,\mathrm{d}x. \tag{178}$$

Consequently S_N is the unique least-squares approximant among all such polynomials.

Proof. Every polynomial of degree at most $2N+1$ is a linear combination of $P_0, \dots, P_{2N+1}$. By construction, up $-S_N$ is orthogonal to each even basis element through P_{2N}; by evenness it is also orthogonal to every odd basis element. Hence

$$\int_{-1}^{1} (\mathrm{up} - S_N)(p - S_N) = 0.$$

Expanding up $-p = (\mathrm{up} - S_N) - (p - S_N)$ gives (178). Equality with the minimum forces the final nonnegative integral to vanish. A polynomial vanishing almost everywhere vanishes identically, so $p = S_N$. □

The one-degree slack is genuine: the odd coefficient of P_{2N+1} is zero. Thus a partial sum visibly of degree $2N$ is already optimal in the larger space of degree at most $2N+1$.

16 Additional summation and spectral identities

16.1 The cosine series on the support

Periodize up with period 2. Since neighboring support intervals meet only at zeros, the periodization agrees with up on $[-1, 1]$. Its Fourier coefficients are $\widehat{\mathrm{up}}(n/2)/2$, so rapid Fourier decay gives the uniformly convergent cosine series

$$\mathrm{up}(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \widehat{\mathrm{up}}(n/2) \cos(\pi n x), \qquad -1 \le x \le 1. \tag{179}$$

Using (38), every coefficient is an explicit infinite product. Many coefficients vanish because one of the sinc factors is evaluated at a nonzero integer.

16.2 Scaled partitions

Equation (43) may be differentiated term by term. For $j \ge 1$,

$$\sum_{k\in\mathbb{Z}} \mathrm{up}^{(j)}\left(t + \frac{k}{m}\right) = 0. \tag{180}$$

The sums are locally finite. Combining this with (51) turns the identity into exact finite cancellations among signed translates of $\mathcal{F}$ at dyadic scales.

16.3 Mean and variance of the Fabius law

From the random-series representation,

$$\mathbb{E}X = \sum_{n\ge 0} \frac{1/2}{2^{n+1}} = \frac{1}{2}. \tag{181}$$

Since $\mathrm{Var}(U_n) = 1/12$ and the coordinates are independent,

$$\mathrm{Var}(X) = \frac{1}{12} \sum_{n\ge 0} 4^{-n-1} = \frac{1}{36}. \tag{182}$$

For $Y = 2X - 1$, this gives $\mathbb{E}Y = 0$ and $\mathrm{Var}(Y) = 1/9$, agreeing with $c_1 = 1/9$. Higher even moments are exactly the c_n of (53).

16.4 An exact self-convolution law

Splitting the random series into its even and odd coordinates gives two independent scaled copies with different weights. At the transform level this is the factorization

$$\widehat{\mathrm{up}}(z) = \widehat{\mathrm{up}}(z/2)\,\mathrm{sinc}(\pi z). \tag{183}$$

In physical space,

$$\mathrm{up}(x) = 2\int_{-1/2}^{1/2} \mathrm{up}(2x-2t)\,\mathrm{d}t = \int_{2x-1}^{2x+1} \mathrm{up}(y)\,\mathrm{d}y, \tag{184}$$

with the normalization understood as convolution with the uniform density on $[-1/2, 1/2]$. Differentiating this integral refinement recovers (21). Conversely, integrating the differential equation and using compact support recovers the convolution law.

17 Computation in exact and numerical regimes

The identities above do not lead to a single universally best numerical method. They lead to a collection of algorithms adapted to different questions. This section makes the distinctions explicit, since confusing an exact arithmetic formula with an endpoint asymptotic is as unhelpful as using a dense dyadic table to evaluate one value at 2^{-10^6}.

17.1 Which representation should be used?

Representation	Principal strength	Principal limitation
Contraction fixed point	A direct constructive proof; after m iterations the uniform error is geometrically small.	Produces approximating functions rather than exact individual values.
Finite convolution / probability	Positive approximants, natural sampling, and a transparent explanation of smoothness.	High resolution requires many convolution levels.
Fourier sinc product	Entire transform, rapid frequency decay, and efficient evaluation away from extreme endpoint cancellation.	Recovering a tiny value of $F(x)$ by oscillatory inversion may lose many digits.
Exact dyadic formula	A closed rational answer and an algebraic proof of representation invariance.	The literal Thue–Morse sum may contain exponentially many terms.
Bit-recursive Taylor evaluator	Exact rational arithmetic with at most one recursive call per set bit of the numerator.	Rational numerators and denominators themselves become very large.
Legendre series	Uniform polynomial approximation on the full closed support, with exact coefficients.	Not specially adapted to resolving the boundary layer at a tiny x.
Lambert saddle expansion	Stable evaluation of $\log F(x)$ at extraordinarily small positive x.	Asymptotic rather than exact; the periodic correction must be retained.

For the fixed-point construction, let T be the transform of (17). If f_0 is any admissible continuous function, then

$$\|T^m f_0 - F\|_\infty \le 2^{-m}\|f_0 - F\|_\infty \le 2^{-m}. \tag{185}$$

Thus the contraction proof itself is a certified numerical scheme. It is excellent for coarse graphs and for constructing a first approximation without any prior arithmetic tables. Exact dyadic computation, however, should use the rational recurrences.

17.2 Precomputing the reciprocal-power table

Put

$$V_n = F(2^{-n}). \tag{186}$$

Eliminating d_n from (62) and (68) gives an especially convenient recurrence involving only earlier V_k:

$$V_0 = 1, \qquad V_{n+1} = \frac{1}{2^{n+1}-1} \sum_{k=0}^{n} \frac{V_k}{2^{\binom{n+1}{2}-\binom{k}{2}}(n+2-k)!}. \tag{187}$$

Every operation is rational. A practical implementation stores the values in an array, reduces each rational after every addition, and reuses factorials and powers of two. This table is precisely the data required by the Taylor blocks in (81).

There are three useful normalization choices.

1. Store V_n directly. This makes the public API simple and is ideal when values are eventually returned as rational numbers.
2. Store $d_n = n!2^{\binom{n}{2}}V_n$. The recurrence has smaller explicit powers of two, and the parity theorem provides a strong integrity check.
3. Store the division-free natural numerators G_n of the half moments. This avoids fraction reduction inside the recurrence but requires restoring a known structured denominator at the end.

The third choice is usually best for bulk generation; the first is usually best for a small library interface.

17.3 Operation count of the bit recursion

Suppose $0 < a < 2^n$, and write the set-bit positions of a in decreasing order as

$$b_1 > b_2 > \cdots > b_h \geq 0, \qquad a = \sum_{j=1}^{h} 2^{b_j}. \tag{188}$$

The evaluator of theorem 9.4 makes exactly $h = \operatorname{popcount}(a)$ nontrivial recursive calls. At the jth call, the Taylor order is $n - b_j$. Hence the number of rational multiply-add steps in straightforward Horner evaluation is

$$\sum_{j=1}^{h}(n-b_j) \leq nh \leq n^2. \tag{189}$$

This is an arithmetic-operation count, not a bit-complexity estimate. The bit lengths of the exact numerator and denominator grow quadratically in n already at $a = 1$, as the factor $2^{\binom{n}{2}}$ in (68) makes unavoidable.

By contrast, the literal formula (75) sums over every $0 \leq h < a$. When a is comparable with 2^n, it therefore has exponentially many outer terms. Its value is conceptual rather than algorithmic: it proves rationality, exposes Thue–Morse cancellation, and connects the answer to q-binomial identities. The Taylor recursion is the compiled form of that cancellation.

Unreduced inputs cause no mathematical difficulty. The exact formula and the executable evaluator satisfy

$$F\left(\frac{2a}{2^{n+1}}\right) = F\left(\frac{a}{2^n}\right). \tag{190}$$

Reducing the pair (a, n) first merely avoids unnecessary work. Signed inputs should be handled at the outermost layer: clamp to 0 for $a \leq 0$, clamp to 1 for $a \geq 2^n$, and call the unit-interval recursion only in the interior.

17.4 Stable endpoint evaluation

For a very small non-dyadic x, direct evaluation of $F(x)$ is the wrong numerical variable. One should compute $\log F(x)$. Set

$$S = -\log x, \qquad L = \log 2. \tag{191}$$

The exact saddle coordinate is the positive solution of

$$L\lambda - \log \lambda = S. \tag{192}$$

Although a library implementation may use the W_{-1} branch directly, Newton iteration is simple and stable:

$$\lambda_{j+1} = \lambda_j - \frac{L\lambda_j - \log \lambda_j - S}{L - \lambda_j^{-1}}. \tag{193}$$

For large S, the starting value

$$\lambda_0 = \frac{S + \log S - \log L}{L} \tag{194}$$

already has small relative error. Once λ is known, evaluate (118) and, when needed, add $A_1(\lambda)/\lambda$ from (142).

The periodic function Ψ should be evaluated by its Fourier series (108). The Gamma factor in (108) decays exponentially with the frequency, while the zeta factor has only moderate growth on the relevant vertical line. Consequently a handful of modes usually suffices for precision far beyond that of ordinary floating-point arithmetic. Since Ψ is real,

$$\Psi(u) = \widehat{\Psi}(0) + 2\Re \sum_{k=1}^{K} \widehat{\Psi}(k) \mathrm{e}^{2\pi i k u} + \text{exponentially small Fourier tail}. \tag{195}$$

Reducing u modulo 1 before evaluating the exponentials avoids loss of phase accuracy.

The saddle expansion and the exact dyadic evaluator overlap over a substantial range. Their agreement is a powerful independent test. For $x = 2^{-n}$, an implementation can compare the exact rational logarithm (computed at high precision) against (164). The difference after the first correction should be $O(n^{-2})$, with the phase evaluated at the exact λ_n, not at n.

17.5 A verification suite suggested by the mathematics

An exact implementation should test identities, not merely decimal snapshots. Particularly effective checks are

$$F(a/2^n) + F((2^n - a)/2^n) = 1, \tag{196}$$

$$F(2a/2^{n+1}) = F(a/2^n), \tag{197}$$

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!), \tag{198}$$

$$F(5/16) = \frac{305857}{2073600}, \tag{199}$$

$$R_n \equiv 1 \pmod 2. \tag{200}$$

The Taylor-block identity can be tested for every a on a modest dyadic grid; the q-binomial expression can be evaluated at several unrelated translations q to test its proved translation independence. For floating-point code, the partition of unity (43), Fourier refinement (39), and agreement between the Legendre and Fourier reconstructions provide complementary global tests.

18 Corrections, conventions, and proof audit

A formal development is most useful when it records not only what is true but also exactly where an attractive informal shortcut ceases to be valid. The following points are the ones most likely to create a silent error in a human treatment.

18.1 Bounded versus signed-global Fabius functions

The bounded function F takes values in $[0, 1]$ and is constant to the right of 1. The signed extension $\mathcal{F}$ is not bounded in that way and instead obeys

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \qquad (x \in \mathbb{R}).$$

They agree only on $[0, 1]$. Thus a statement such as "$F'(x) = 2F(2x)$ for all real x" is correct only when F denotes the signed extension. The explicit Thue–Morse series (11) makes the distinction unavoidable and proves that the global sum is locally finite rather than merely conditionally convergent.

Likewise, the support statement for up concerns the topological support: $\operatorname{supp}\operatorname{up} = [-1, 1]$, even though $\operatorname{up}(\pm 1) = 0$. Endpoint formulas must distinguish closed support from the open set on which the function is positive.

18.2 The Reshetnikov normalization

Equation (74) contains the factor $2^{+\binom{n-1}{2}}$. The negative exponent printed in the abstract of one version of the arithmetic paper is incompatible with its own equation (27) and with the integrality theorem. The positive exponent is the one proved by the exact recurrence. The resulting first values are

$$R_1, R_2, \ldots, R_7 = 1,\ 5,\ 15,\ 1001,\ 5985,\ 2853675,\ 26261235, \tag{201}$$

all odd as predicted.

The strongest common-denominator formula in the arithmetic paper depends on a separate divisibility conjecture. The formal development keeps the unconditional multiplier bound and the conditional exact denominator statement logically separate. A computational match for the first many levels is evidence for the conjecture, not a proof.

18.3 The local asymptotic derivation

Write the logarithmic endpoint profile in a dyadic variable as

$$g(t) = \log F(2^{-t}). \tag{202}$$

A proposed elementary derivation substitutes a main term $G(t)$ into the exact difference-differential relation. Two steps in that derivation cannot be made for an unspecified remainder $R = g - G$:

1. a coefficient comparison drops the actual difference $R(t) - R(t - 1)$;
2. a later estimate uses a bound on $R'(t)$ that has not been assumed or proved.

Even the explicit main term has a residual of order

$$\left(\frac{\log t}{t}\right)^2, \tag{203}$$

with a nonzero leading coefficient; it is not $O(t^{-2})$. Therefore the printed iteration does not establish the claimed bounded periodic remainder. The rigorous route is the exact negative-Laplace decomposition of section 10, followed by Bromwich inversion and a quantitative saddle analysis.

There is also a small but conceptually useful conversion issue. Substituting $x = 2^{-t}$ in the literal elementary small-x expression does not give the separately displayed dyadic expression term for term. Their difference is

$$\frac{(\log\log 2)^2}{2(\log 2)^2\, t}, \tag{204}$$

which is $O(t^{-1})$. It is harmless for a coarser theorem but must be retained when the display itself claims accuracy through the inverse-logarithmic term.

18.4 Why the periodic term cannot be averaged away

Replacing $\Psi(\lambda)$ by its mean changes the logarithm by a bounded quantity, so it preserves only very coarse statements such as (160). It does not preserve a sharp multiplicative asymptotic. Indeed, for every nonzero integer k,

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{\log 2} \neq 0.$$

The nonvanishing follows from the absence of zeros of Γ, the fact that $\zeta(s) \neq 0$ on $\Re s = 1$ away from its pole, and $\chi_k \neq 0$. Hence Ψ is nonconstant. Choosing two fractional saddle phases where it takes different values produces two sequences along which the ratio to an uncorrected main term has different limits. This is a structural obstruction, not a matter of choosing a better constant.

18.5 A visually good quotient of exponentials has the wrong boundary scale

A proposed elementary approximation to the displaced bump has the form

$$Q(x) = \frac{1 + \exp(-|x|^{-\alpha})}{1 + \exp((1-2|x|)/(x^2 - |x|))}, \qquad \alpha = \frac{3}{4}\sqrt{2\pi}. \tag{205}$$

It can look strikingly accurate on a plot. Put $y = x + 1$, the distance from the left support endpoint. For $0 < y \leq 1/4$ one has the rigorous bound

$$Q(-1+y) \leq 2\exp\left(-\frac{1}{2y}\right). \tag{206}$$

But theorem 14.2 shows

$$\exp(-c/y) = o(F(y)) \qquad (y \downarrow 0) \tag{207}$$

for every $c > 0$. Therefore

$$Q(-1+y) = o(F(y)), \tag{208}$$

so the quotient is not asymptotically equivalent at the endpoint. This is a useful warning: a graph on a linear scale is almost blind to the distinction between $\exp[-\Theta((\log(1/y))^2)]$ and $\exp[-\Theta(1/y)]$ once both values are tiny.

18.6 What the formalization proves, and what it deliberately does not claim

The public aggregate contains exact proofs of the two source papers' named results, together with the additional dyadic, q-binomial, Legendre, and sharp-asymptotic theorems described here. It does not promote a source conjecture to a theorem, does not infer derivative bounds for an unspecified asymptotic remainder, and does not identify two global functions merely because they coincide on $[0, 1]$. These negative design decisions are part of the mathematics: they preserve the hypotheses needed by every later theorem.

19 Conclusion

The Fabius function is unusual not because it has one exotic property, but because several mathematical structures coincide in one object.

- A contraction on symmetric distribution functions constructs it and proves uniqueness.
- An infinite sum of independent uniform variables explains positivity, symmetry, and smoothing.
- Rvachev's fold is a compactly supported C^∞ density satisfying an exact functional-differential equation.
- Its Fourier transform is an entire sinc product with superalgebraic real-axis decay.
- Thue–Morse splitting creates a signed global solution and exact formulas for every derivative.
- Rational moment recurrences evaluate every dyadic argument exactly.
- A Taylor recursion turns the closed Thue–Morse formula into an efficient bit-sensitive algorithm.
- A negative-Laplace product and a lower-Lambert saddle expose the true endpoint scale, its Gamma–zeta periodic fluctuation, and its complete asymptotic expansion.
- A uniformly convergent Legendre series converts the same exact dyadic data into globally optimal polynomial approximants on the support.

The dyadic and asymptotic theories are not separate curiosities. The exact values $F(2^{-n})$ determine the moments and the Legendre coefficients; their logarithmic growth is governed by the same saddle geometry that controls arbitrary small arguments. The periodic term is the analytic shadow of dyadic self-similarity, while the Thue–Morse signs encode the same binary splitting algebraically. Seen together, these descriptions turn a famous "pathological example" into a remarkably coherent piece of analysis.

A Exact data

The following tables collect enough exact values to reproduce the examples in the text and to seed independent implementations.

A.1 Moment and half-moment sequences

n	c_n	d_n
0	1	1
1	$\frac{1}{9}$	$\frac{1}{2}$

n	c_n	d_n
2	$\frac{19}{675}$	$\frac{5}{18}$
3	$\frac{583}{59535}$	$\frac{1}{6}$
4	$\frac{132809}{32531625}$	$\frac{143}{1350}$
5	$\frac{46840699}{24405225075}$	$\frac{19}{270}$
6	$\frac{4068990560161}{4133856862760625}$	$\frac{1153}{23814}$
7	—	$\frac{583}{17010}$

The c_n column satisfies (53); the d_n column satisfies (62). Substitution in (68) gives the next table.

A.2 Reciprocal powers and normalized odd integers

n	$F(2^{-n})$	$v_2(F(2^{-n}))$	R_n
0	1	0	—
1	$\frac{1}{2}$	-1	1
2	$\frac{5}{72}$	-3	5
3	$\frac{1}{288}$	-5	15
4	$\frac{143}{2073600}$	-10	1001
5	$\frac{19}{33177600}$	-15	5985
6	$\frac{1153}{561842749440}$	-22	2853675
7	$\frac{583}{179789679820800}$	-29	26261235

The valuation column is computed from (72); it is not obtained by factoring the displayed decimal-sized denominators. This is exactly the kind of redundancy that makes the table useful as a test vector.

B Proof dependency and Lean module guide

The Lean development is intentionally split into small files, while this article groups results by mathematical theme. The following map gives the principal entry points. File names are relative to

`Analysis/FabiusFunction/Lean/FabiusFunction/`.

Mathematical layer	Principal module(s)	Role in the proof
Exact sequences	`Arithmetic.lean`	Defines binary weight, Thue–Morse signs, c_n, d_n, division-free numerators, reciprocal values, closed dyadic formulas, and executable evaluators over $\mathbb{Q}$.

Mathematical layer	Principal module(s)	Role in the proof
Analytic interface	`Basic.lean`	Separates bounded F, compactly supported up, and signed $\mathcal{F}$; defines Fourier products and moment integrals.
Existence and uniqueness	`Existence.lean`	Constructs F as the fixed point of a strict contraction and proves uniqueness from dense dyadic values.
Differential equations	`Differential.lean`	Proves evenness and the global Rvachev equation, including the gluing argument at the fold point, then bootstraps to C^∞.
Original characterization	`OriginalCharacterization.lean`, `OriginalUniqueness.lean`	Formalizes the compact-support characterization and proves uniqueness through the Fourier refinement product.
Probability	`ProbabilityRepresentation.lean`	Builds the infinite product probability space, proves independence, identifies the random series CDF with F, and obtains up as its density.
Finite convolutions	`ConvolutionApproximation.lean`, `StepApproximant.lean`	Connects polynomial densities, repeated convolutions, weak convergence, and the step approximants of the original paper.
Fourier and Poisson	`Fourier.lean`, `Poisson.lean`	Proves the sinc product, inversion, rapid decay, Poisson summation, cosine series, and partitions of unity.
Signed extension	`GlobalExtension.lean`	Proves local finiteness of the Thue–Morse translate sum, the global differential equation, and exact iterated-derivative formulas.
Moment arithmetic	`MomentPowerSeries.lean`, `HalfMomentGenerating.lean`, `Paper06487.lean`	Bridges the exact recurrences to integrals and entire generating functions and packages the arithmetic paper's named statements.
Closed dyadic form	`DyadicCorrectness.lean`, `DyadicClosedForm.lean`	Proves affine Thue–Morse cancellation, the Taylor-block identity, and agreement of the fast evaluator with the closed formula.
Global dyadic values	`GlobalDyadic.lean`	Extends exact computation beyond the unit interval using triangular reduction and Thue–Morse signs.
q-binomial formulas	`FabiusQBinomialTaylor.lean`, `FabiusDyadicQBinomialScalar.lean`	Proves the finite q-binomial forms, arbitrary translation invariance, scalar extension, and dyadic-representation invariance.
Negative-Laplace product	`NegativeLaplace.lean`, `PeriodicCorrection.lean`	Splits the logarithmic product into its quadratic drift, periodic finite part, and exponentially small tail.
Fourier analysis of Ψ	`PeriodicFourier.lean`, `PeriodicMean.lean`, `FabiusSharpConstant.lean`	Computes all Fourier coefficients and the mean of the periodic correction and identifies the sharp Euler–Stieltjes constant.
Lambert saddle	`FabiusLambertSaddle.lean`, `QuantitativeSaddle.lean`, `FabiusSharpAsymptotic.lean`	Introduces the lower-Lambert coordinate, performs exact Bromwich reduction, and proves the quantitative leading asymptotic.
Higher asymptotics	`FabiusFirstSaddleCorrection.lean`, `FabiusFullAsymptoticExpansion.lean`, `FabiusWikipediaExpansion.lean`	Derives the first periodic correction, every higher coefficient, the elementary log–log conversion, and the refutation of the uncorrected claimed error.

Mathematical layer	Principal module(s)	Role in the proof
Decay comparisons	`FabiusDecayComparison.lean`, `FabiusQuotientExponentialMismatch.lean`	Places F between algebraic and ordinary flat-exponential scales and proves the endpoint failure of the visual quotient model.
Legendre expansion	`FabiusLegendreCoefficients.lean`, `FabiusLegendreSeries.lean`	Computes exact coefficients, proves endpoint-uniform absolute convergence, and establishes least-squares optimality.
Public aggregates	`Paper05442.lean`, `Paper06487.lean`, `PaperStatements.lean`	Expose source-facing theorem names and the paper-coverage interface from the public import `FabiusFunction`.

C Notation crosswalk

Different sources use F, φ, θ, and up for overlapping objects. The following crosswalk records the convention used here.

This article	Definition	Frequent source notation
$F(x)$	Bounded CDF on all of $\mathbb{R}$, constant 0 to the left and 1 to the right.	Fabius function on $[0,1]$; sometimes the same letter is reused for the global extension.
$\mathrm{up}(x)$	Even compactly supported density $F(1-\lvert x\rvert)$ on $[-1,1]$.	$\mathrm{up}(x)$ or $\varphi(x)$.
$\mathcal{F}(x)$	Signed Thue–Morse extension, zero on $(-\infty,0]$ and satisfying $\mathcal{F}' = 2\mathcal{F}\circ(2\,\cdot)$.	$F(x)$ or $\theta(x)$ in papers that work globally.
c_n	Even moment $\int x^{2n}\,\mathrm{up}(x)\,\mathrm{d}x$.	c_n in the arithmetic paper.
d_n	Half moment $n\int_0^1 x^{n-1}\,\mathrm{up}(x)\,\mathrm{d}x$, with $d_0=1$.	d_n in the arithmetic paper.
Ψ	Centered or explicitly mean-separated one-periodic correction in the negative-Laplace finite part.	Often absent from elementary accounts.
λ	Exact positive solution of $\lambda 2^{-\lambda} = x$.	Lower-Lambert saddle coordinate $-W_{-1}(-x\log 2)/\log 2$.

When comparing formulas, one should first identify which global convention is in force and only then compare signs, supports, and endpoint values.

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